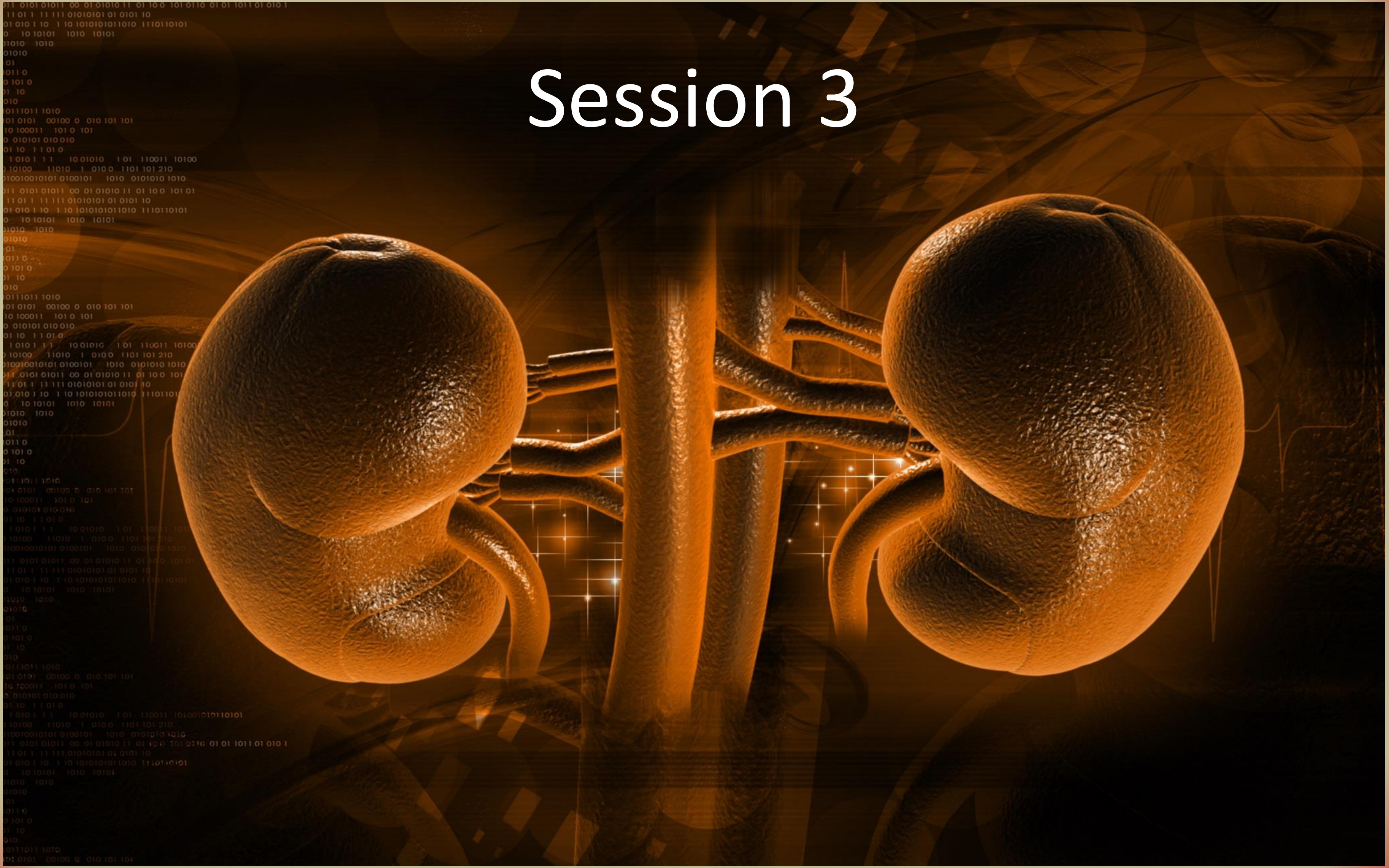
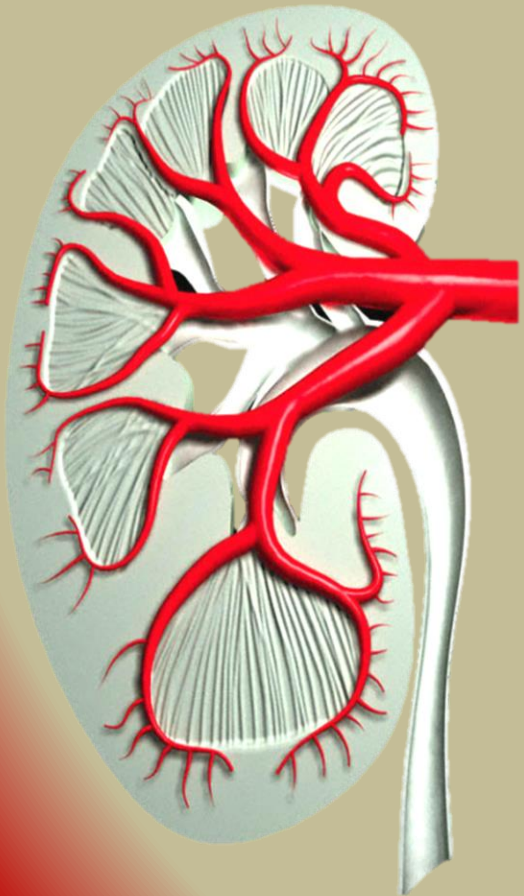
Session 3



The Urinary System



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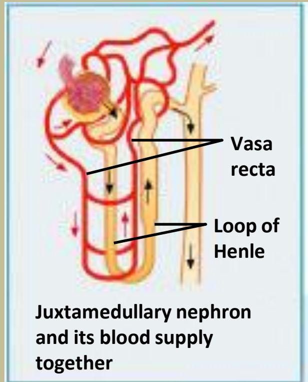
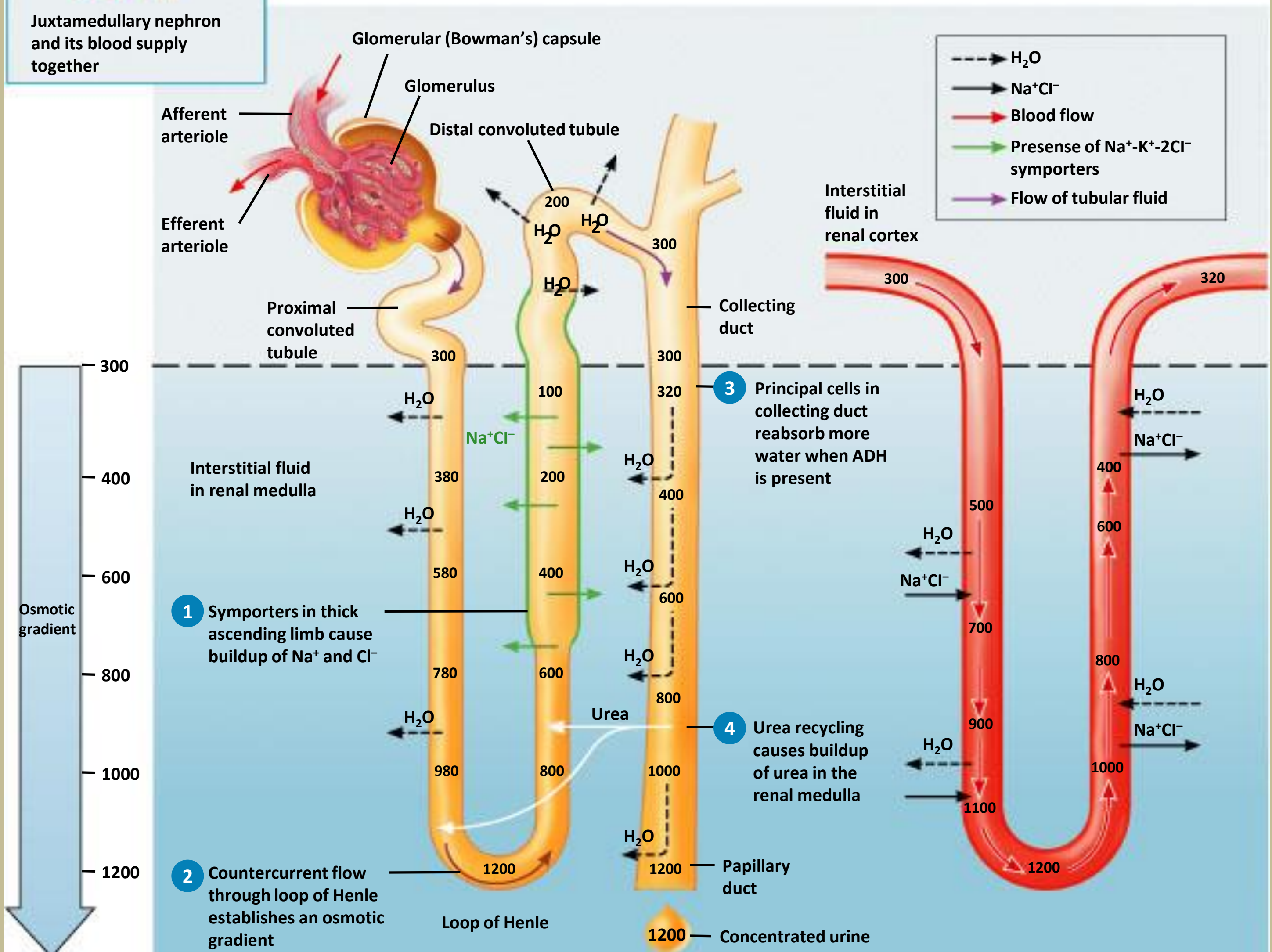
PHYSIOLOGY OF
THE RENAL SYSTEM

Mechanism of urine concentration in long-loop juxtamedullary nephrons





(a) Reabsorption of Na^+Cl^- and water in a long-loop juxtamedullary nephron

(b) Recycling of salts and urea in the vasa recta

Summary of filtration, reabsorption, and secretion in the nephron and collecting duct

PROXIMAL CONVOLUTED TUBULE

Reabsorption (into blood) of filtered:

Water	65% (osmosis)
Na^+	65% (sodium-potassium pumps, symporters, antiporters)
K^+	65% (diffusion)
Glucose	100% (symporters and facilitated diffusion)
Amino acids	100% (symporters and facilitated diffusion)
Cl^-	50% (diffusion)
HCO_3^-	80–90% (facilitated diffusion)
Urea	50% (diffusion)
Ca^{2+}, Mg^{2+}	variable (diffusion)

Secretion (into urine) of:

H^+	variable (antiporters)
NH_4^+	variable, increases in acidosis (antiporters)
Urea	variable (diffusion)
Creatinine	small amount

At end of PCT, tubular fluid is still isotonic to blood (300 mOsm/liter).

LOOP OF HENLE

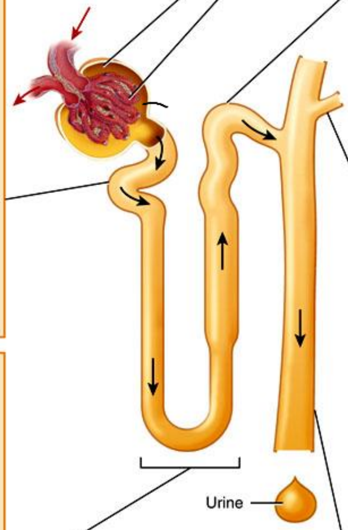
Reabsorption (into blood) of:

Water	15% (osmosis in descending limb)
Na^+	20–30% (symporters in ascending limb)
K^+	20–30% (symporters in ascending limb)
Cl^-	35% (symporters in ascending limb)
HCO_3^-	10–20% (facilitated diffusion)
Ca^{2+}, Mg^{2+}	variable (diffusion)

Secretion (into urine) of:

Urea	variable (recycling from collecting duct)

At end of loop of Henle, tubular fluid is hypotonic (100–150 mOsm/liter).



RENAL CORPUSCLE

Glomerular filtration rate: 105–125 mL/min of fluid that is isotonic to blood

Filtered substances: water and all solutes present in blood (except proteins) including ions, glucose, amino acids, creatinine, uric acid

EARLY DISTAL CONVOLUTED TUBULE

Reabsorption (into blood) of:

Water	10–15% (osmosis)
Na^+	5% (symporters)
Cl^-	5% (symporters)
Ca^{2+}	variable (stimulated by parathyroid hormone)

LATE DISTAL CONVOLUTED TUBULE AND COLLECTING DUCT

Reabsorption (into blood) of:

Water	5–9% (insertion of water channels stimulated by ADH)
Na^+	1–4% (sodium-potassium pumps and sodium channels stimulated by aldosterone)
HCO_3^-	variable amount, depends on H^+ secretion (antiporters)
Urea	variable (recycling to loop of Henle)

Secretion (into urine) of:

K^+	variable amount to adjust for dietary intake (leakage channels)
H^+	variable amounts to maintain acid-base homeostasis (H^+ pumps)

Tubular fluid leaving the collecting duct is dilute when ADH level is low and concentrated when ADH level is high.

Figure 26.20 Tortora - PAP 12/e

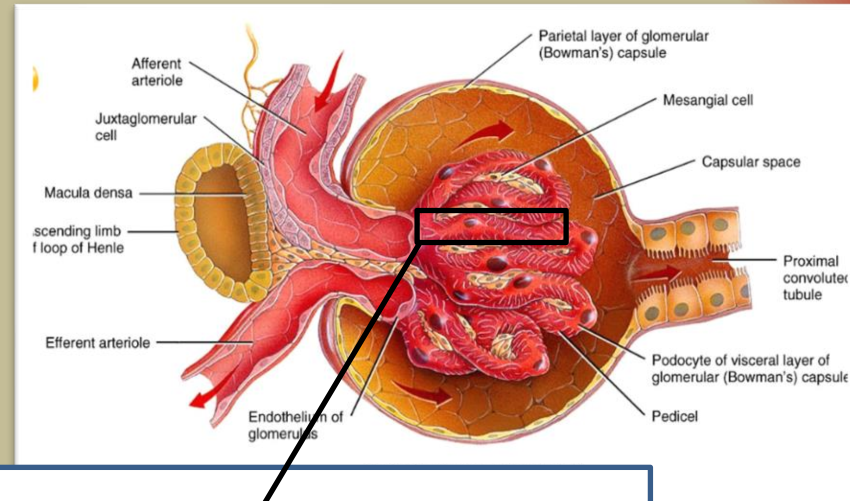
Glomerular filtration

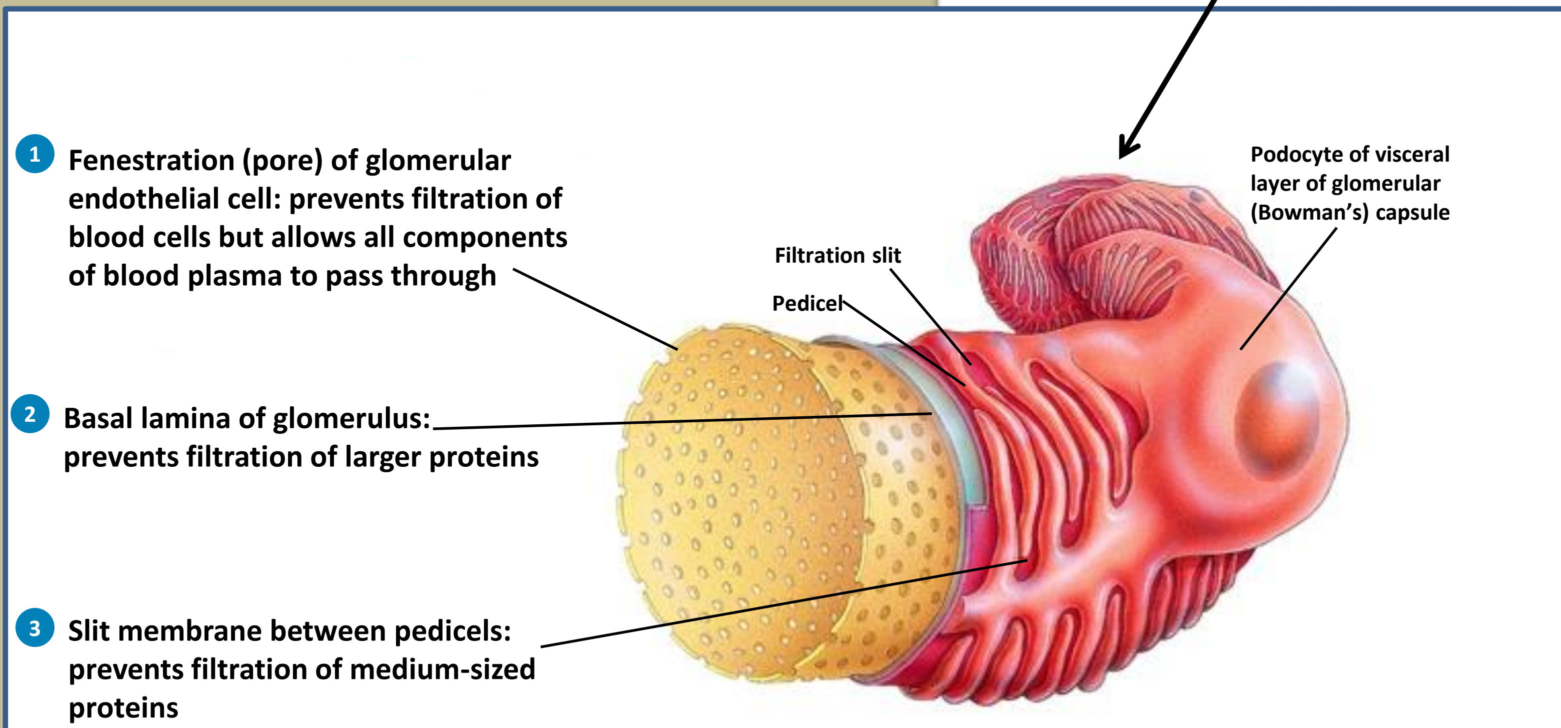
- Glomerular filtrate: fluid that enters the capsular space
  - Daily volume: 150-180 liters
  - More than 99% returned to blood plasma via tubular reabsorption

- Filtration barrier
  - Permits filtration of water and small solutes
  - Prevents filtration of most plasma proteins, blood cells and platelets
  - 3 barriers to cross:
    - glomerular endothelial cells fenestrations,
    - basal lamina between endothelium
    - Podocytes and pedicels of podocytes create filtration slits
  - Volume of fluid filtered is large because of large surface area, thin and porous membrane, and high glomerular capillary blood pressure

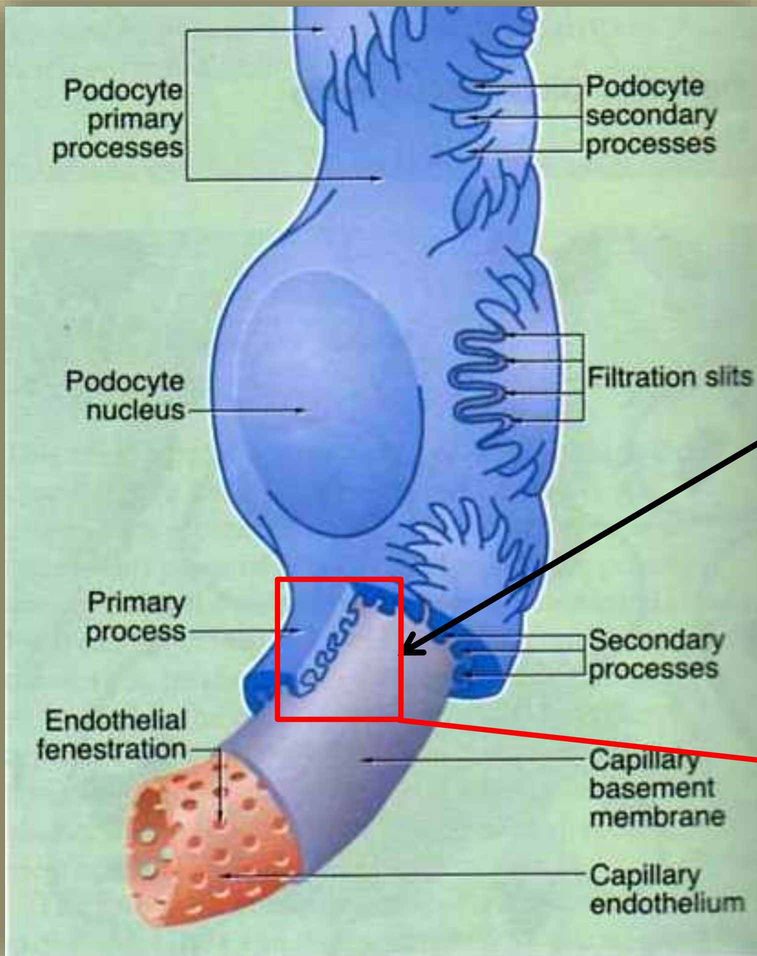
Glomerulus fly-through

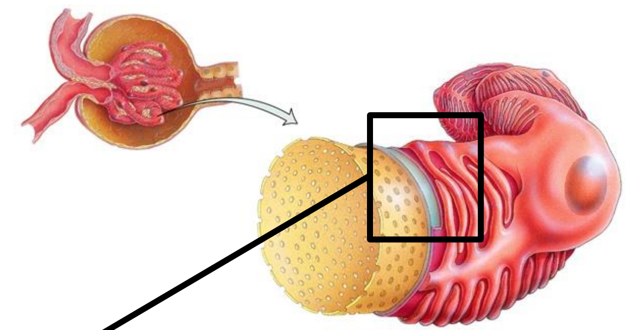
The Filtration membrane



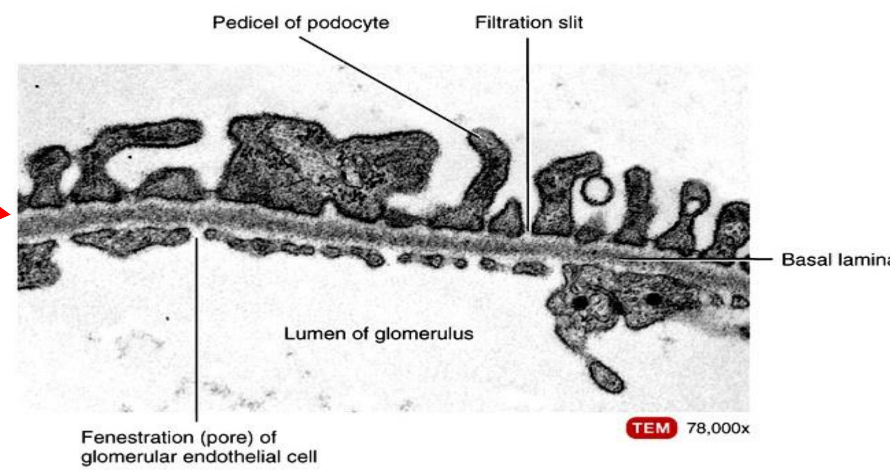


(a) Details of filtration membrane



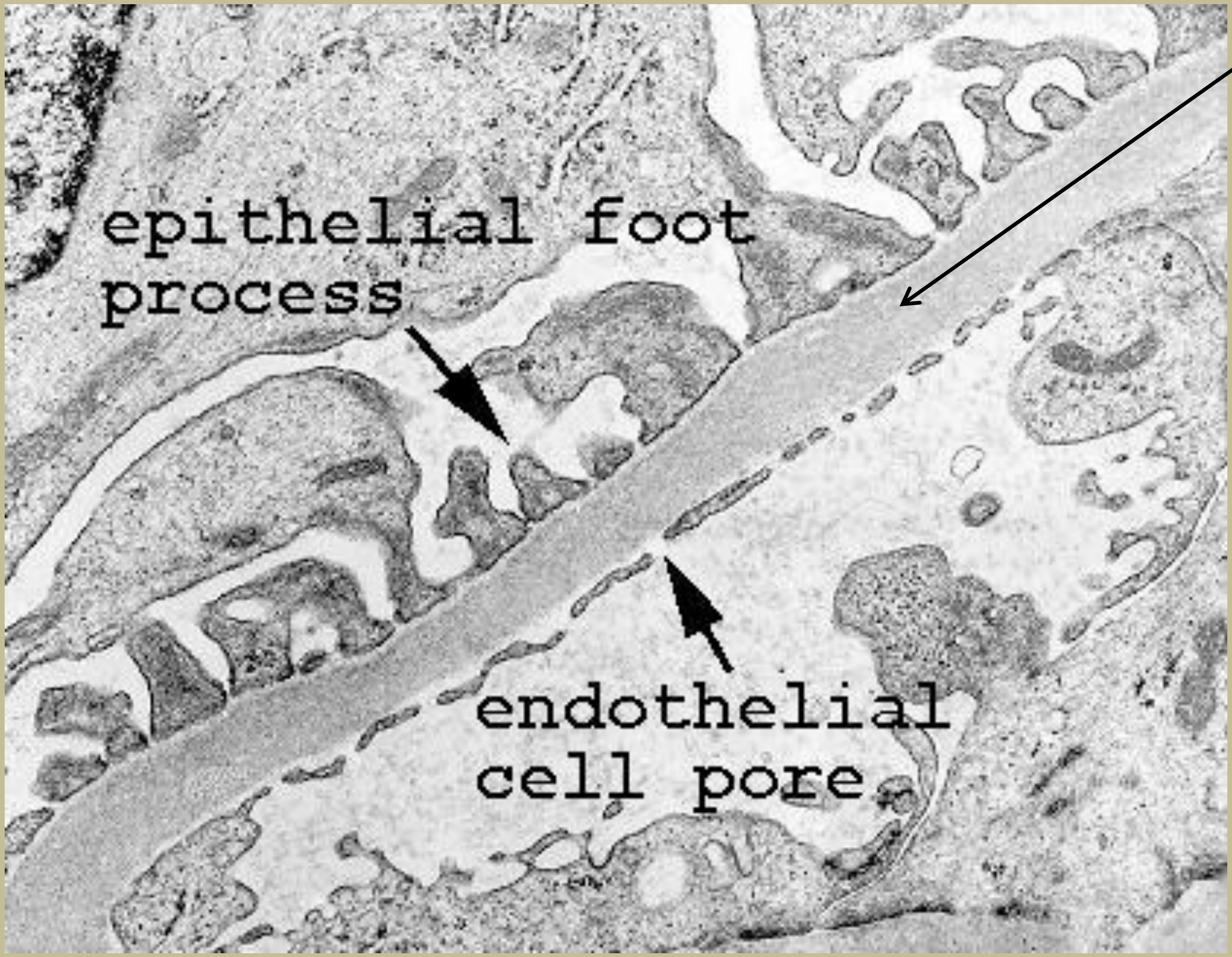


The filtration membrane



(b) Filtration membrane

Figure 26.08 Tortora - PHA 11/e



Basal membrane

Structure of the Basal membrane

Important in relation to the pathology of the kidney:

- Nephrotic syndrome, nephritic syndrome, etc.

The glomerular basal lamina:

≈ 300 nm thick, made up of 3 layers

1. Lamina rara externa
2. Lamina densa in the middle
3. Lamina rara interna

Lamina densa

Type IV collagen and laminin, which form a network,

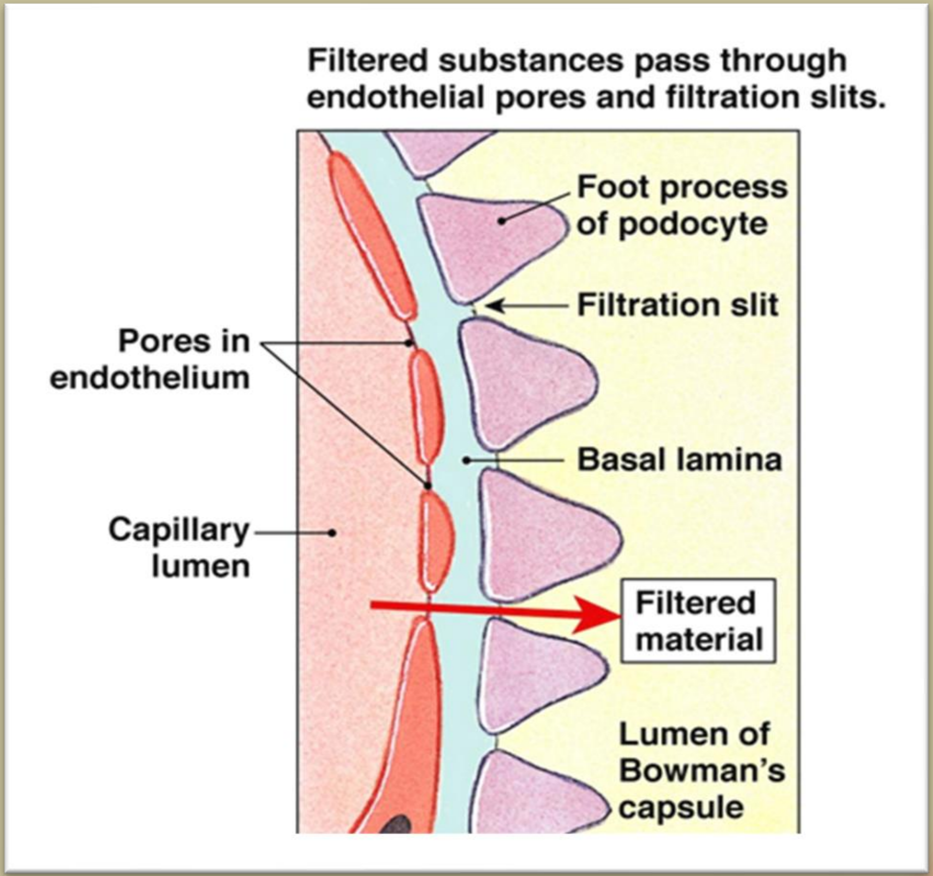
The network acts as a filter

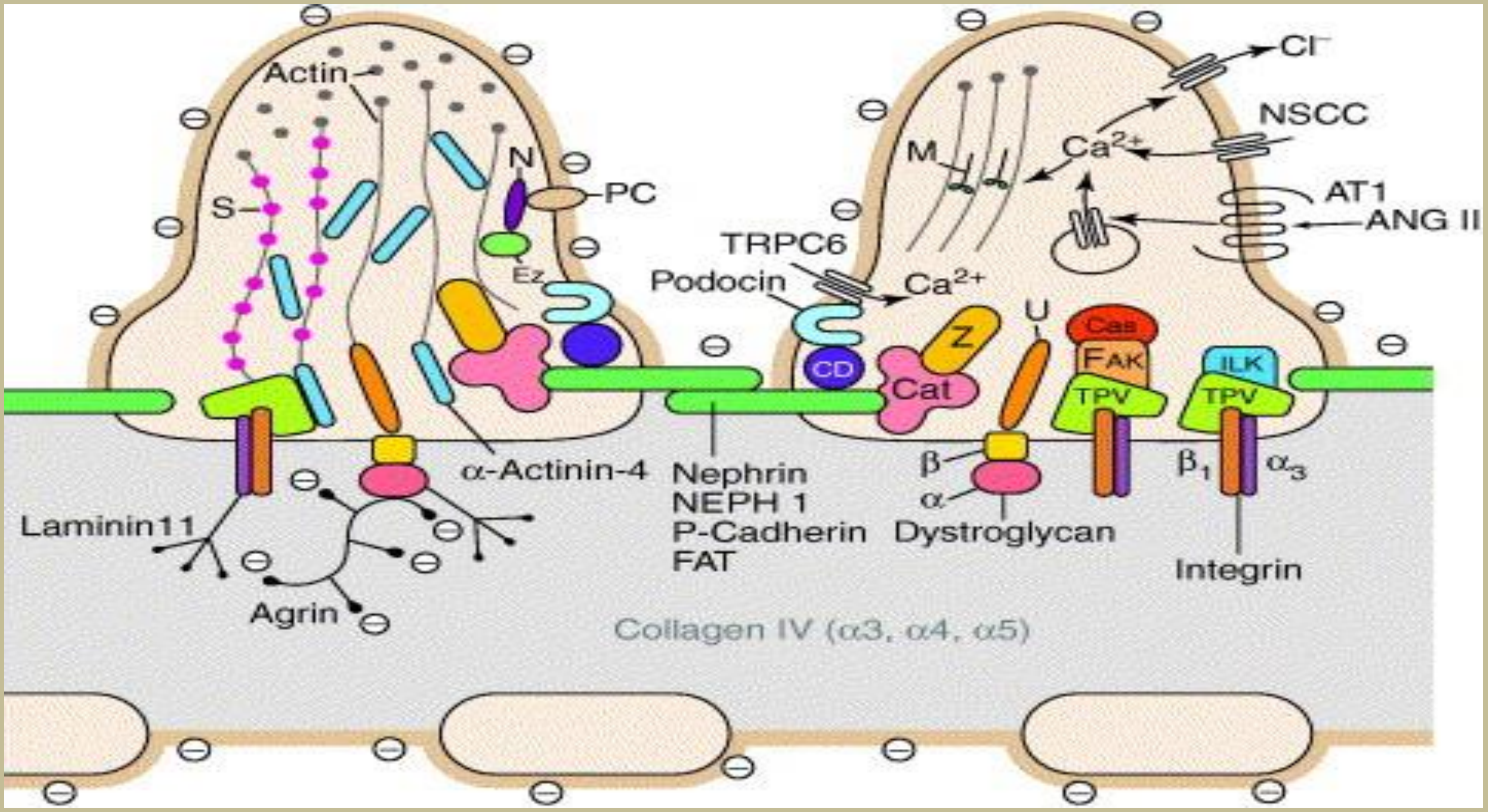
Lamina rara

There are large amounts of heparan sulfate, strong negative charge

Glomerular Filtration

- The *capsular epithelium (podocytes)* surrounding the outer surface of the capillary basement membrane forms a part of the *glomerular filtration membrane.*





Glomerular filtration barrier

Two podocyte foot processes bridged by the slit membrane, the glomerular basement membrane (GBM) and the porous capillary endothelium are shown.

Wilhelm Kriz
TRPC6 ? a new podocyte gene involved in focal segmental glomerulosclerosis

Trends in Molecular Medicine Volume 11, Issue 12 2005 527 – 530
http://dx.doi.org/10.1016/j.molmed.2005.10.001

Glomerular Filtration

a complex process but a simple outcome

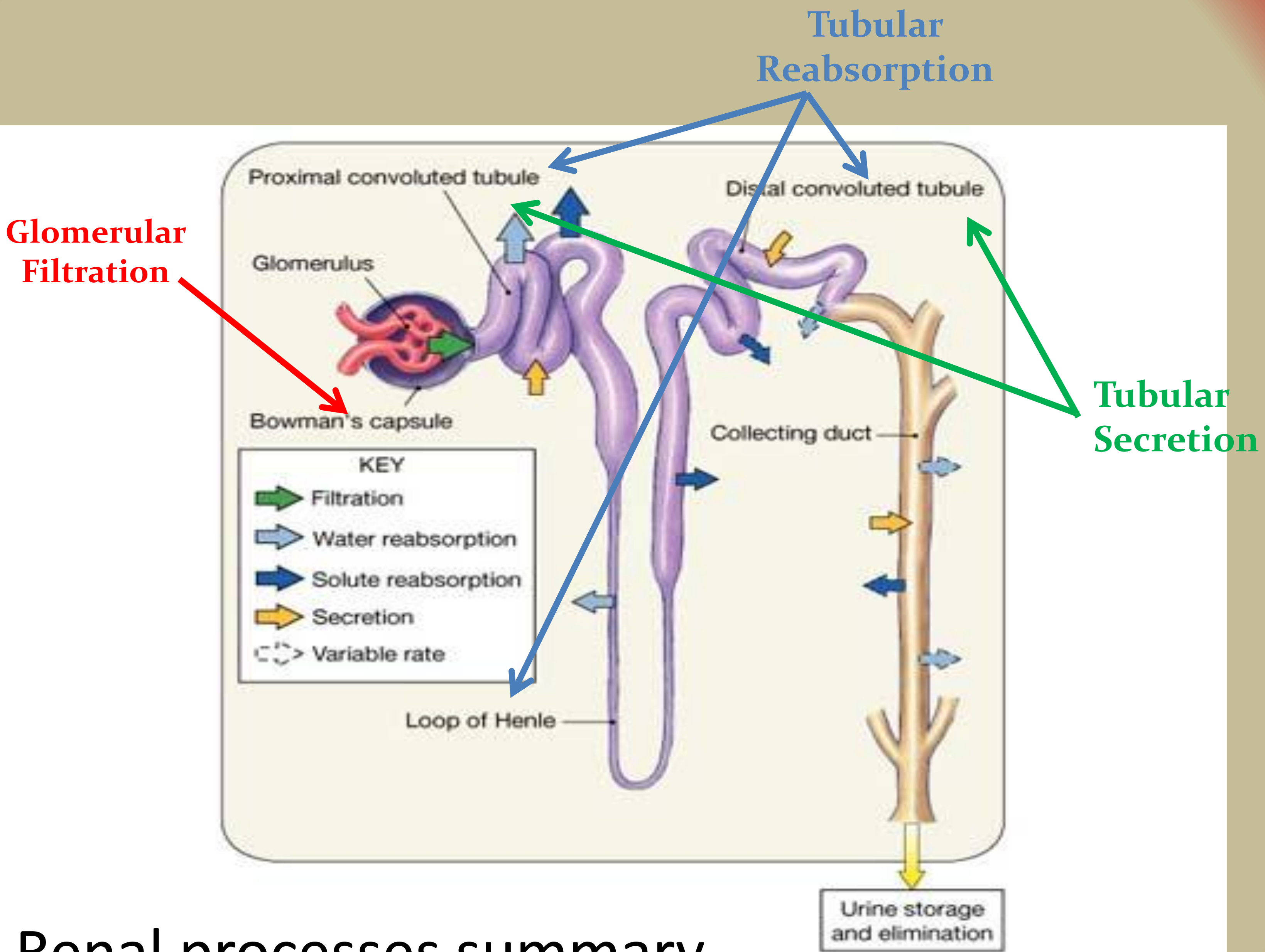
- Small molecules such as *water, glucose, and ionic salts* *except for plasma proteins* are able to pass through the slits and form an **ultra filtrate**.

- Podocytes are also involved in regulation of glomerular filtration rate (GFR).
    - When podocytes contract, they cause closure of filtration slits.

↓

Decreases the *GFR*

(by reducing the surface area available for filtration)



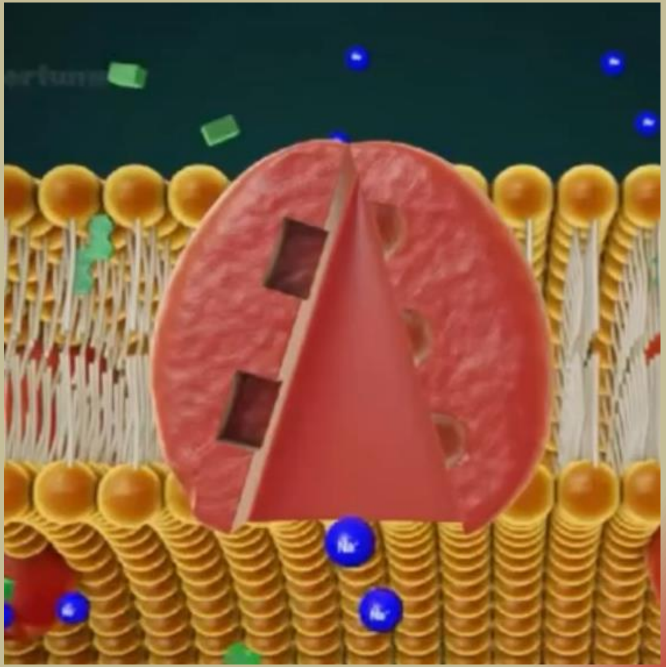
Renal processes summary

Reabsorption routes and transport mechanisms

- Reabsorption routes
  - Paracellular reabsorption
    - Between adjacent tubule cells
    - Tight junction do not completely seal off interstitial fluid from tubule fluid
    - Passive
  - Transcellular reabsorption – through an individual cell

- Transport mechanisms

  Reabsorption of Na^+ especially important
  - Primary active transport
    - Sodium-potassium pumps (in basolateral membrane only)
  - Secondary active transport
    - Symporters, antiporters



Reabsorption routes:
paracellular reabsorption and **transcellular reabsorption**

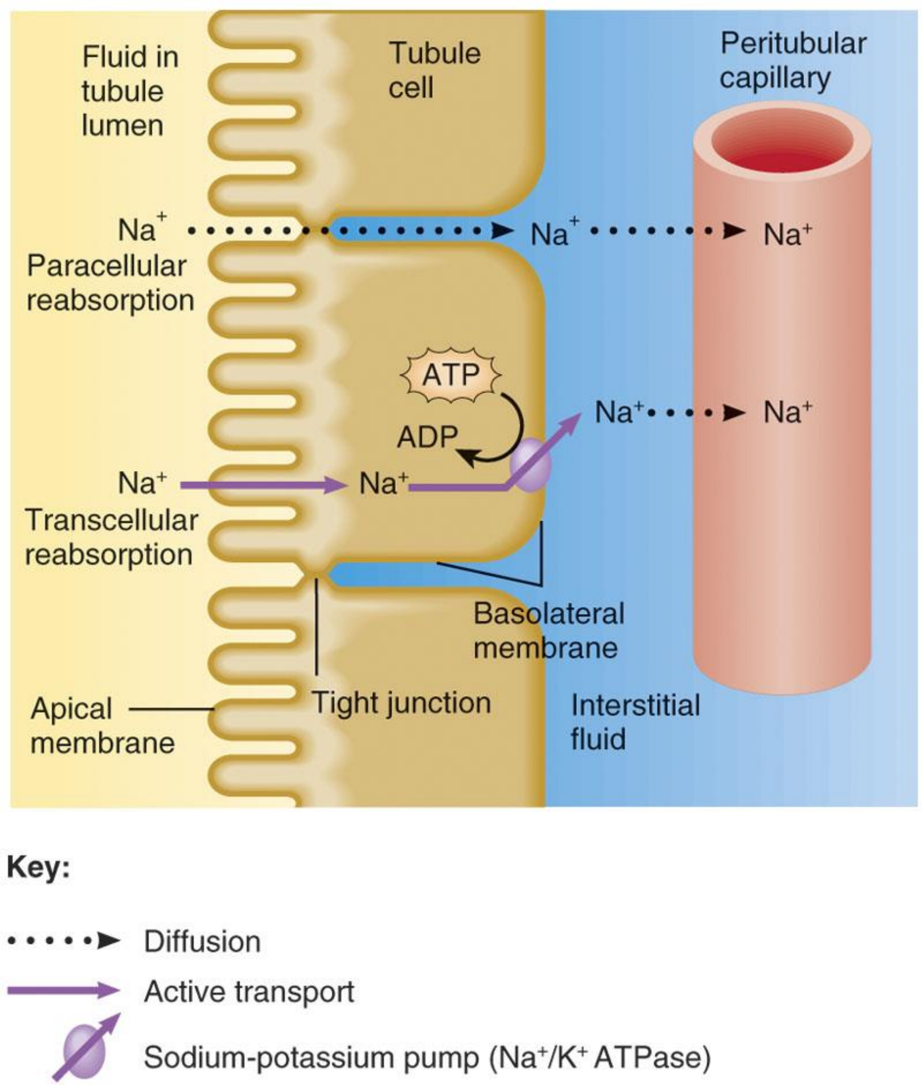
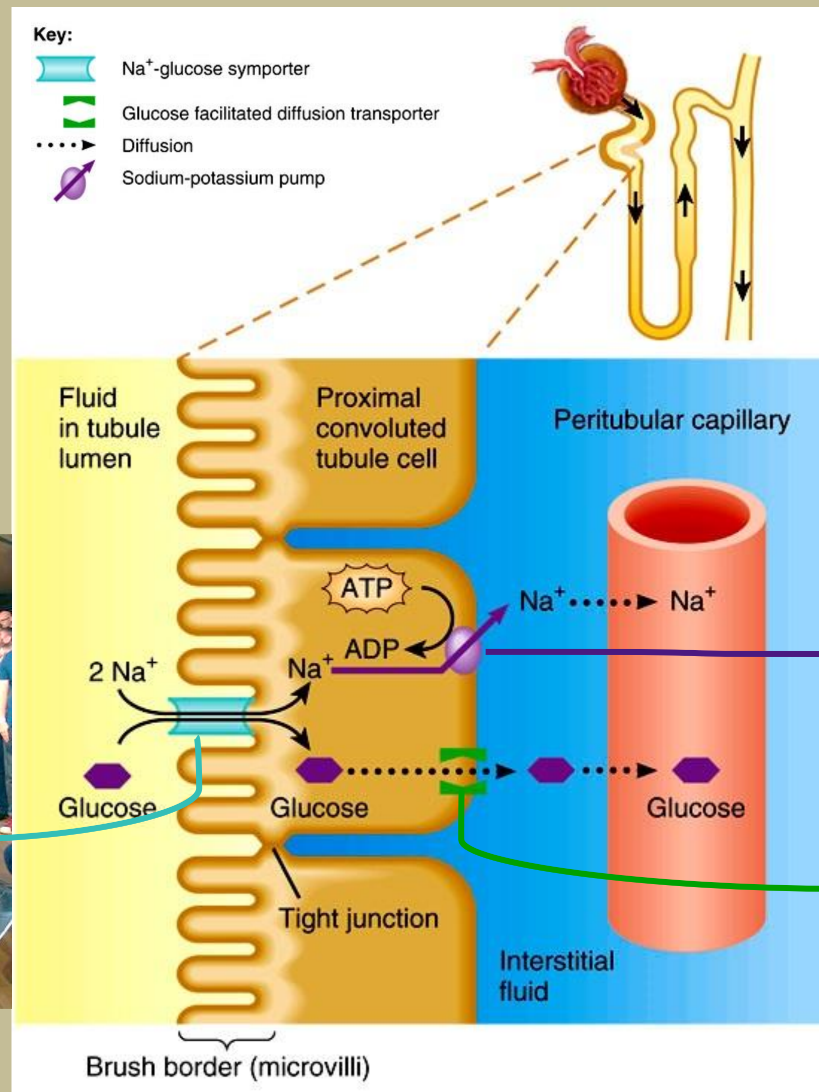
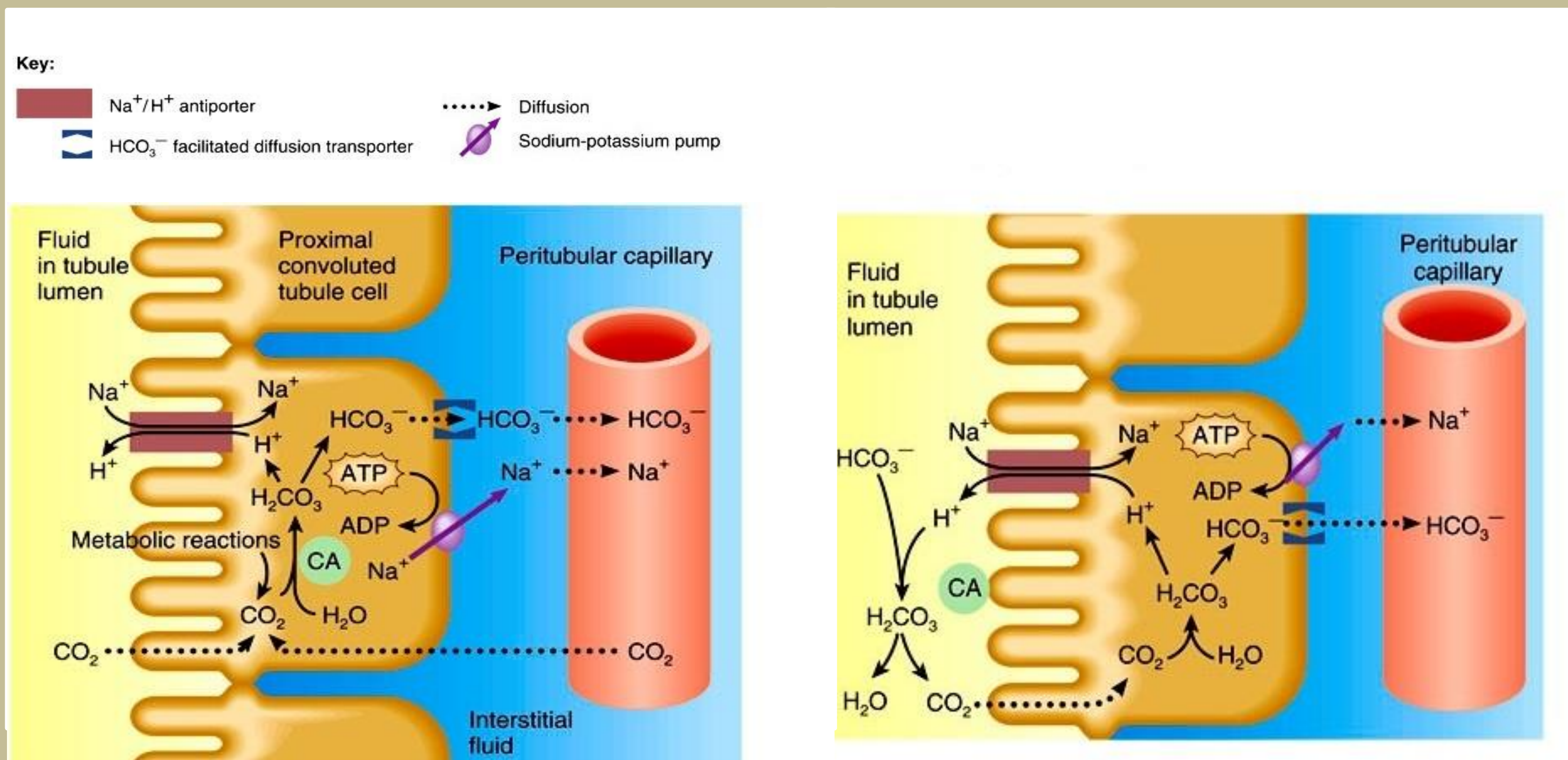


Figure 26.11 Tortora - PAP 12/e

Reabsorption and secretion in the proximal convoluted tubule



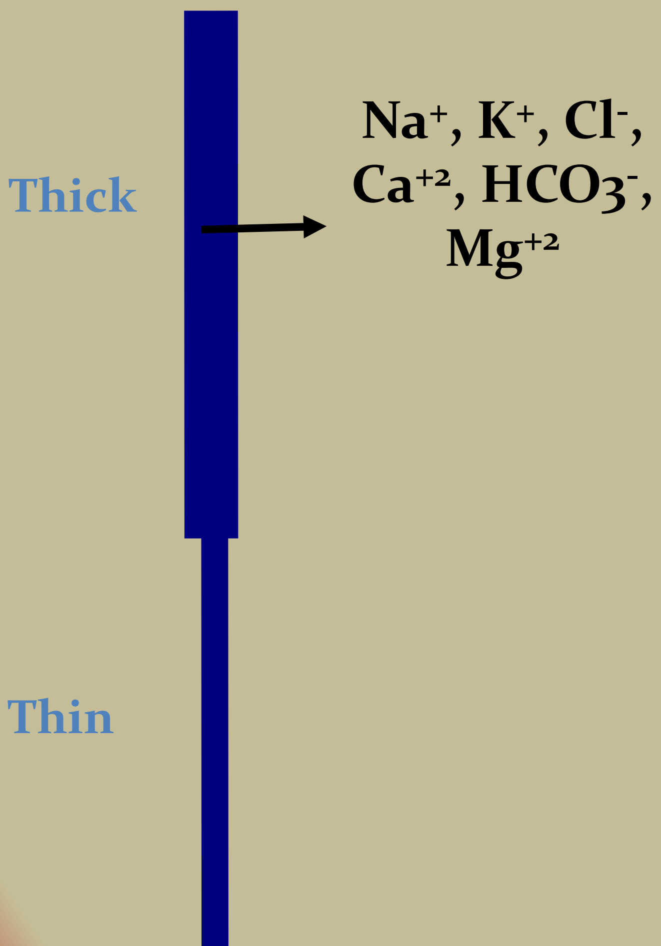
Reabsorption and secretion in the proximal convoluted tubule

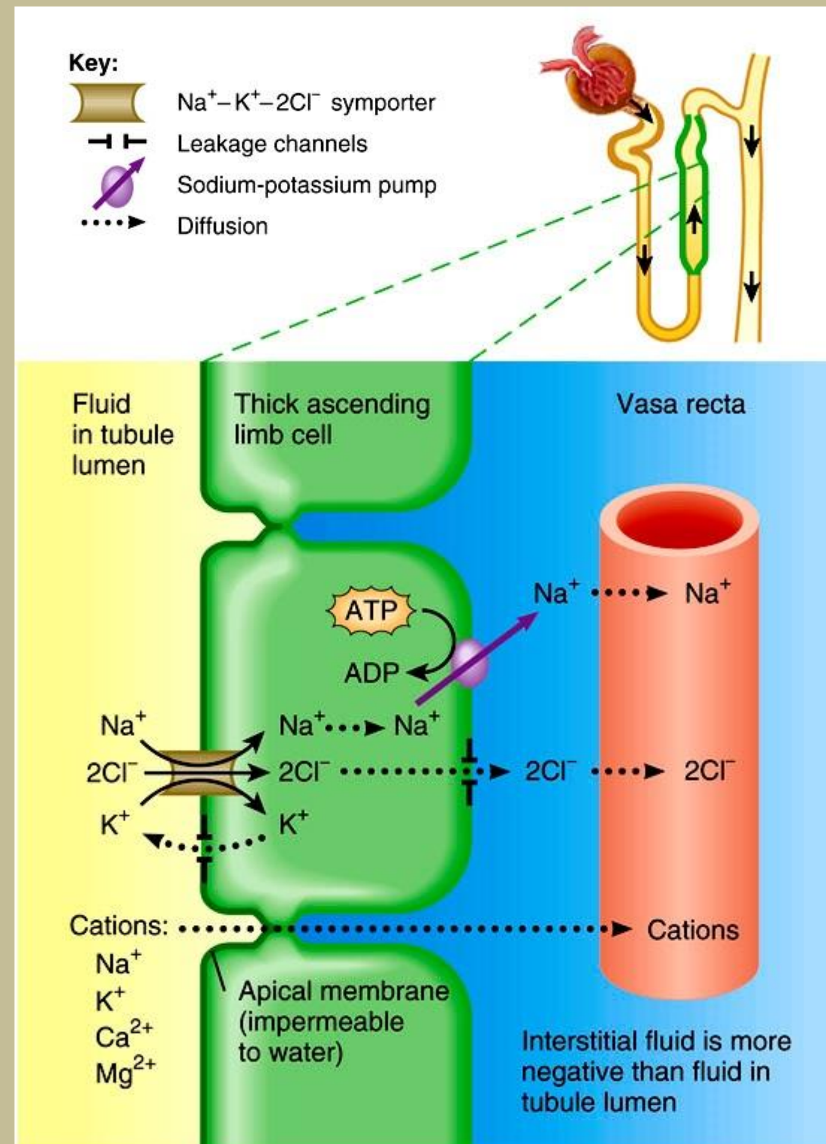


(a) Na^+ reabsorption and H^+ secretion

(b) HCO_3^- reabsorption

Na^+–K^+-$2Cl^-$ symporter in the thick ascending limb of the loop of Henle





Hormonal regulation of tubular reabsorption and secretion

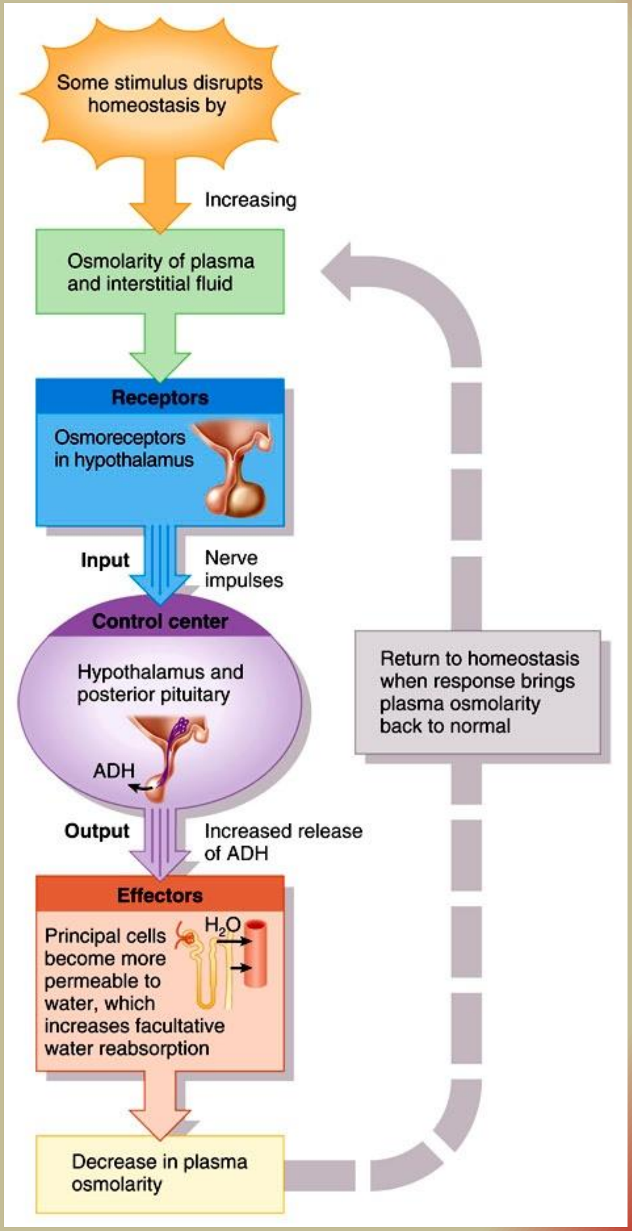
When blood volume and blood pressure decrease:

- **Angiotensin II** -
  - Decreases GFR, enhances reabsorption of Na^+, Cl^- and water in PCT
- **Aldosterone**
  - Stimulates principal cells in collecting duct to reabsorb more Na^+ and Cl^- and secrete more K^+
- **Parathyroid hormone**
  - Stimulates cells in DCT to reabsorb more Ca^{2+}

Regulation of facultative water reabsorption by ADH & ANP

- **Antidiuretic hormone**
  (ADH or vasopressin)
  - Increases water permeability of cells by inserting aquaporin-2 in last part of DCT and collecting duct

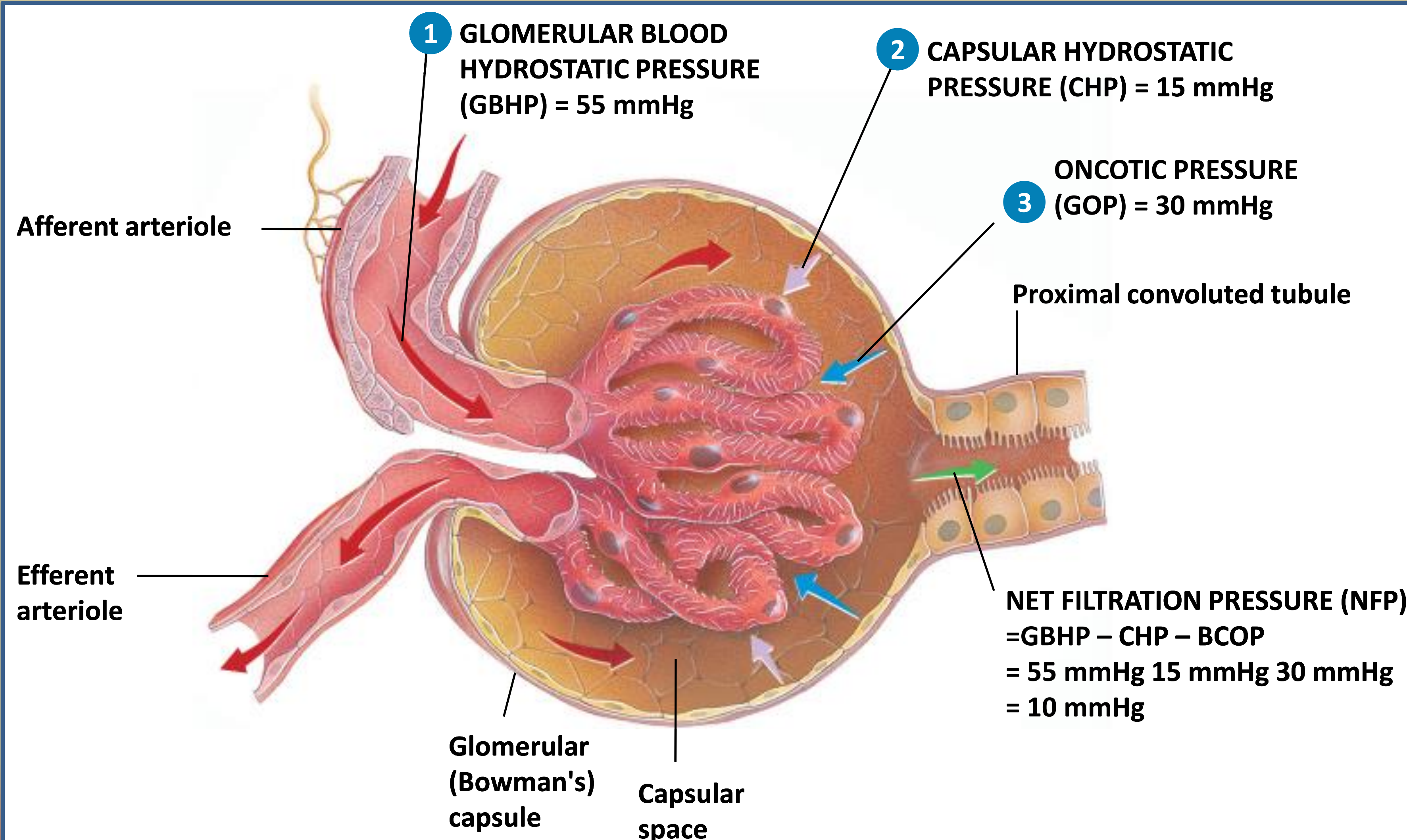
- **Atrial natriuretic peptide (ANP)**
  - Large increase in blood volume promotes release of ANP
  - Decreases blood volume and pressure by:
    - inhibiting reabsorption of Na^+ and water in PCT and collecting duct,
    - suppress secretion of ADH and aldosterone



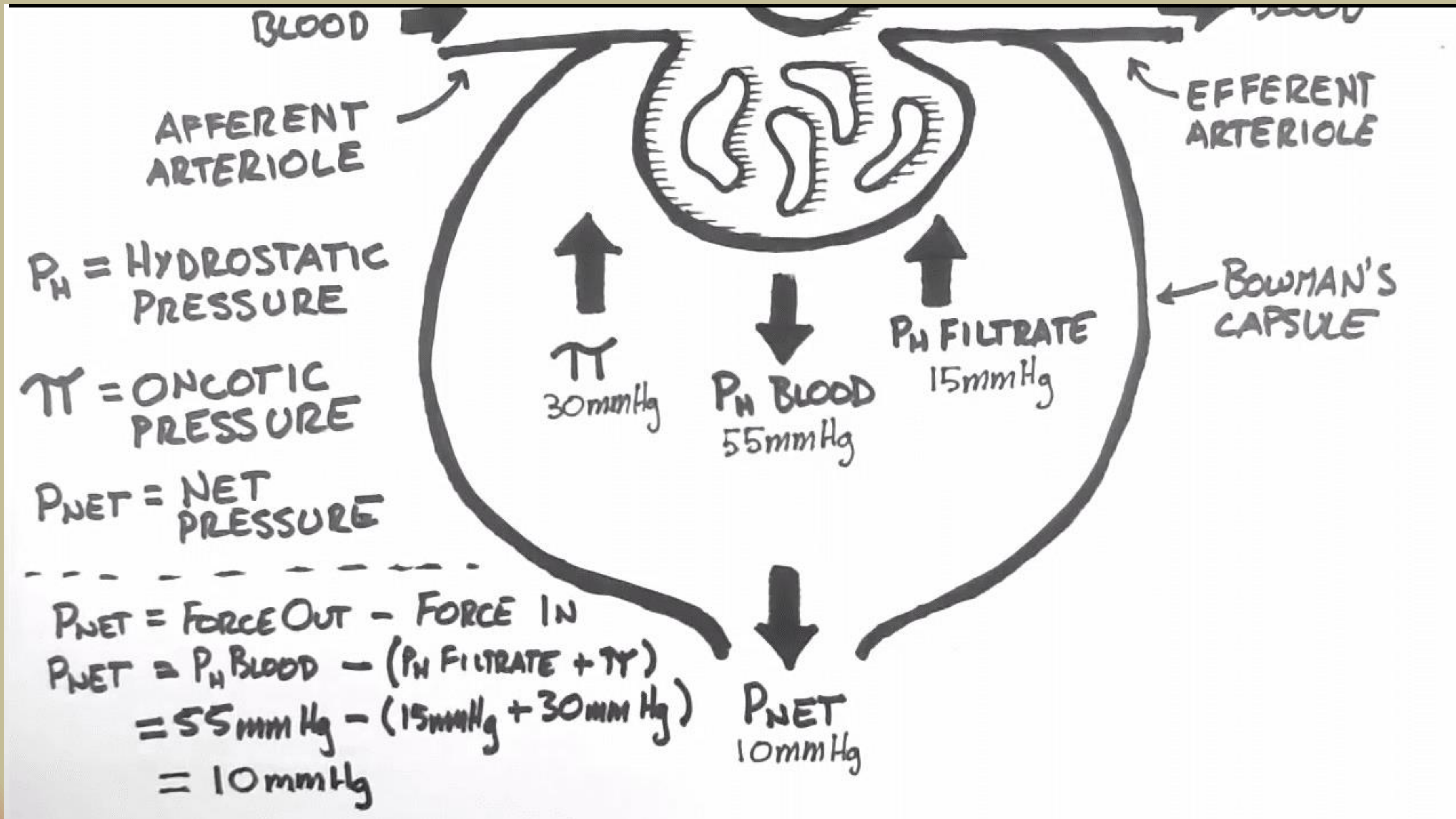
Substance	% of filtrate reabsorbed	Mechanisms
Salt and water	approximately two-thirds (60 -70 %)	Passively across the luminal membrane via transcellular transport, which is then actively reabsorbed across the basolateral membrane via the Na/K/ATPase pump.
Organic solutes (primarily glucose and amino acids)and inorganic phosphate	100%	Reabsorbed via secondary active transport through cotransport channels driven by the sodium gradient out of the PCT epithelial cells.
Potassium (K+)	approximately 65%	Reabsorbed by paracellular mechanisms.

Substance	% of filtrate reabsorbed	Mechanisms
Urea	approximately 50%	As water leaves the lumen, the concentration of urea increases which facilitates diffusion in the late proximal tubule.
Phosphate (HPO42-)	approximately 80%	Reabsorbed via secondary active transport mechanisms and exit across the basolateral membrane by facilitated diffusion.
Chloride (Cl-)	Less than 65%	The higher concentration of chloride in the late PCT favors simple diffusion from the tubular lumen into the renal interstitium through tight junctions.
Bicarbonate (HCO3-)	approximately 80%	Via counter transport mechanisms.

The pressures that drive glomerular filtration



Forces of filtration



Net filtration pressure

- Net filtration pressure (NFP) is the total pressure that promotes filtration

NFP = GBHP – CHP – GOP

  - Glomerular blood hydrostatic pressure (GBHP) is the blood pressure of the glomerular capillaries forcing water and solutes through filtration slits
  - Capsular (filtrate) hydrostatic pressure (CHP) is the hydrostatic pressure exerted against the filtration membrane by fluid already in the capsular space and represents “back pressure”
  - Glomerular Oncotic Pressure (GOP) due to presence of proteins in blood plasma and also opposes filtration. Sometimes also known as *Blood colloid osmotic pressure (BCOP).*

Glomerular filtration rate (GFR)

- Glomerular filtration rate:
  - amount of filtrate formed in all the renal corpuscles of both kidneys each minute

$$GFR = \frac{\text{Urine Concentration} \times \text{Urine Flow}}{\text{Plasma Concentration}}$$

  - Homeostasis requires kidneys to maintain a relatively constant GFR
    - Too high – substances pass too quickly and are not reabsorbed
    - Too low – nearly all reabsorbed and some waste products not adequately excreted
  - GFR directly related to pressures that determine net filtration pressure

Coherence between GFR and filtration

- GFR reports on kidney function
  - Normally 125 ml/min
  - Can change due to illness or because of age

- Filtration rate value of a substance can be calculated as
  - GFR x $P_{[substance]}$ ($P_{[substance]=}$ concentration of substance in plasma)

- Excretion rate as:
  - Vu x $U_{[substance]}$ (Vu = urine concentration; U = urine volume per minute)

- For all substances are true:

  Excretion = (filtration + secretion) – absorption

Clearance and GFR

- **Clearance:** Characterizes the kidney 's ability to secrete a substance

  Definition: Plasma volume, which is cleaned completely of substance per time unit

- Excreted amount of substance is related to its plasma concentration

  C = (urine volume/min x urine concentration) / plasma concentration

  = (U * V) / P = X ml/min

  That is the amount of plasma that is cleaned of the substance per/min

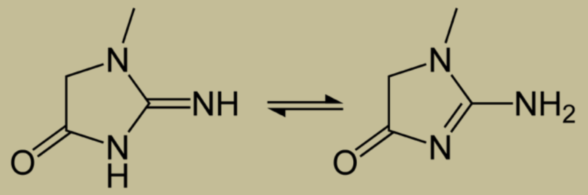
- Clearance is an objective of GFR, if the substance to be measured on is:
  - Freely filterable in the glomerulus (such as water)
  - Is not protein-bound
  - Is not absorbed or secreted into the tubules
  - To assess renal clearance, the system must be in equilibrium

Creatinine

- In clinical practice, serum creatinine clearance is used to measure GFR.
- An energy source for muscle cells
  - Phosphocreatin + ADP ↔ ATP + Creatinine
- is proportional to the muscle mass
  - people with large muscle mass have a high creatinine
- 90%, filtered, secreted 10% → Creatinine clearance overestimates GFR
- Cockcroft-Gault equation for estimated creatinine clearance:

$$eC_{Cr} = \frac{(140 \text{ - Age}) \times \text{Mass (in kilograms)} \times Constant}{\text{Serum Creatinine (in } \mu\text{mol/L)}}$$

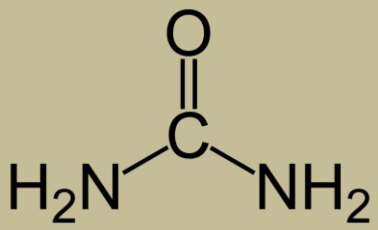
- constant = 1.23 (men) or 1.04 (women)
- Abnormal renal clearance indicates kidney disease



Wikipedia

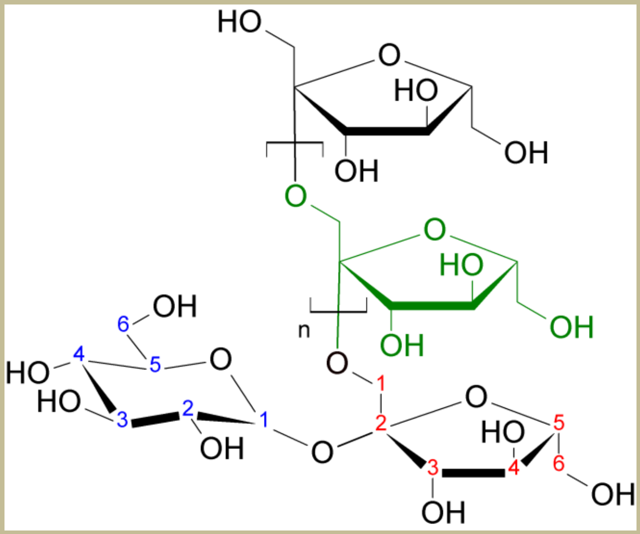
BUN (urea)

- Protein Degradation
- Formation depending on the nutrition
- Filter 90%, 10% reabsorbed
- BUN clearance underestimates GFR



Wikipedia

Inulin



- Polysaccharide
- Not reabsorbed and filtered 100%
- Can be used to estimate GFR

- Not used due to clinical limitationn

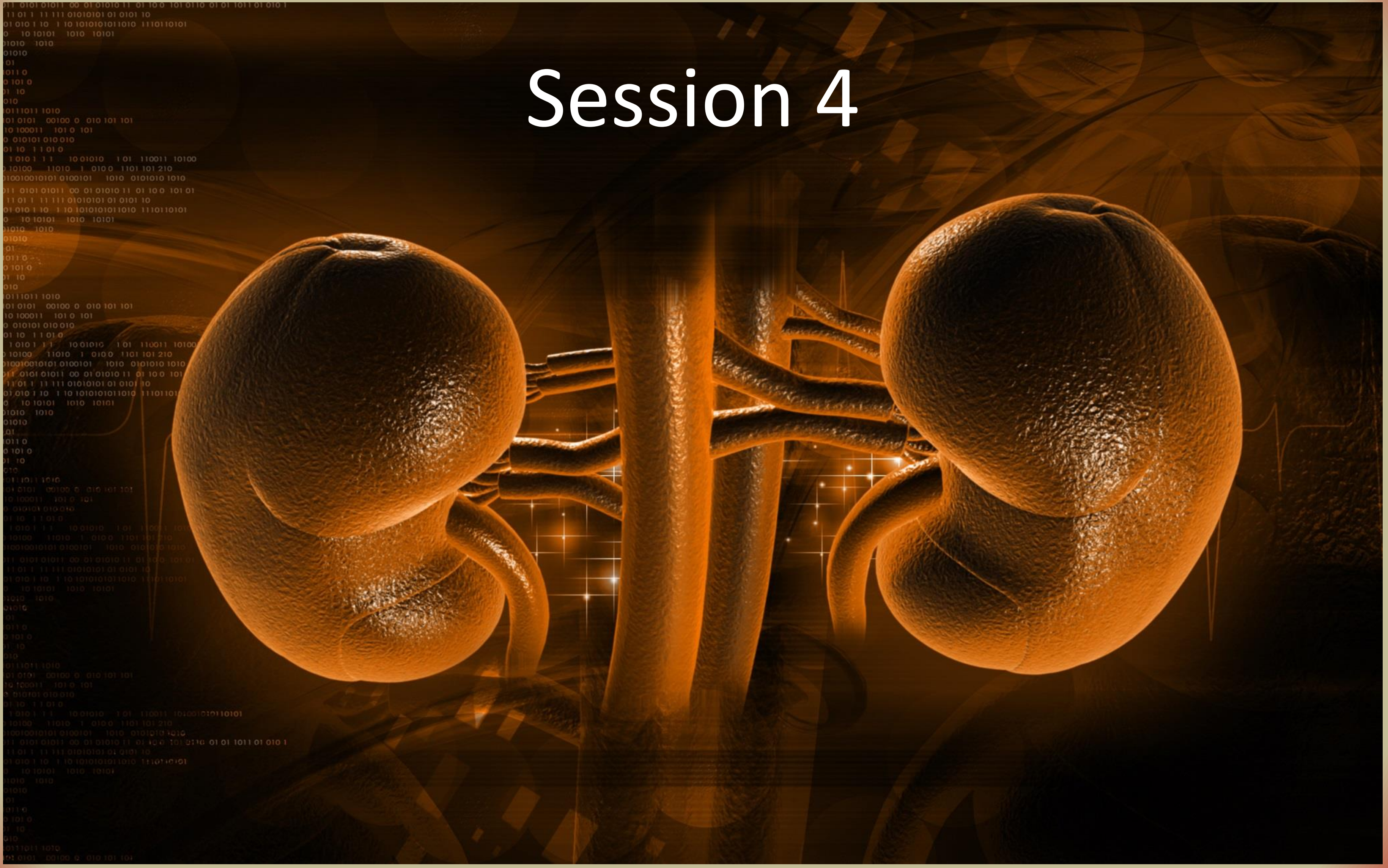
Chromium EDTA

- If creatinine clearance is problematic because of the urine collection inaccuracies.

- You can use Chromium EDTA GFR to determine if accurate impairment provision is necessary.

- The investigation is carried out by the injection of the radioactive isotope chromium EDTA, and the patient is followed for several hours with blood sampling.

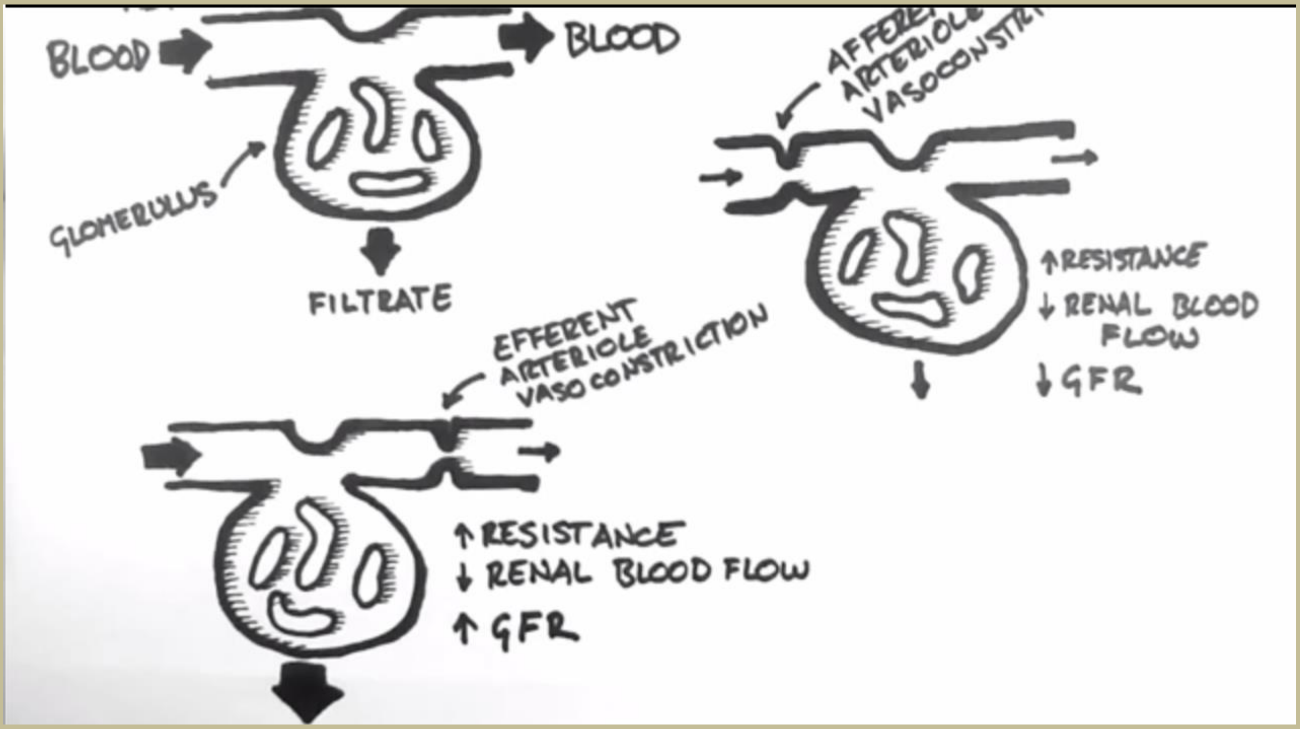
Session 4



Autoregulation of GFR

- Normally 180 L filtrate (pre-urine) is formed daily in capillaries which means 125 ml/minute
- This depends on hydrostatic pressure (Among other factors)
- Hydrostatic pressure is dependent on blood pressure
- Kidneys themselves maintain constant renal blood flow and GFR

Vasoconstriction and GFR



Renal blood flow (RBF)

- The amount of blood that flows through kidney
  - Is proportional to the renal plasma flow (RPF, 600 ml/min)
    - RBF = RPF/(1-Hct).   Hct = hematocrit, red blood cells in the blood

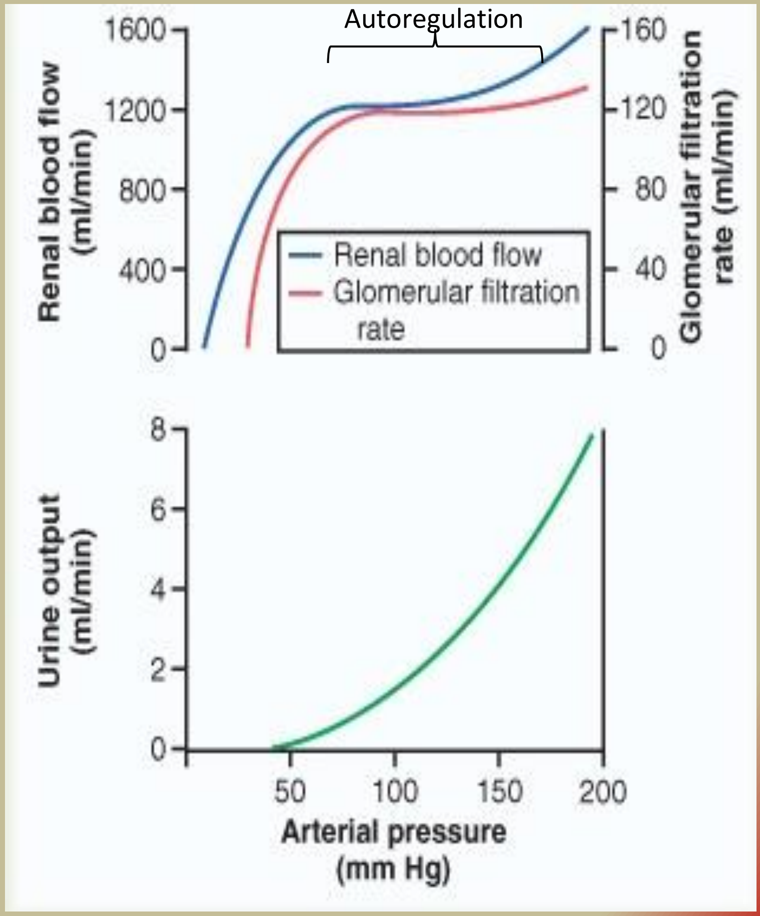
- RBF can be estimated from the clearance of PAH
  - Para-aminohippuric acid (PAH) is cleaned from the kidney by one passage

$$RPF = C_{PAH} \quad \rightarrow \quad RBF = C_{PAH}/(1\text{-}Hct)$$

Blood flow and filtration

Correlation between GFR auto-regulation and blood pressure in capillaries:

- Effect of vasoconstriction
  - **Afferent**
    - (GFR -, RBF -)
  - **Efferent**
    - (GFR +, RBF -)



Mechanisms regulating GFR

1. Renal auto-regulation
2. Neural regulation
3. Hormonal regulation

Mechanisms regulating GFR

1. Renal auto-regulation

- Myogenic mechanism:

  contraction of smooth muscle cells in afferent arterioles due to the opening of "stretch-activated Ca^{++} channels"

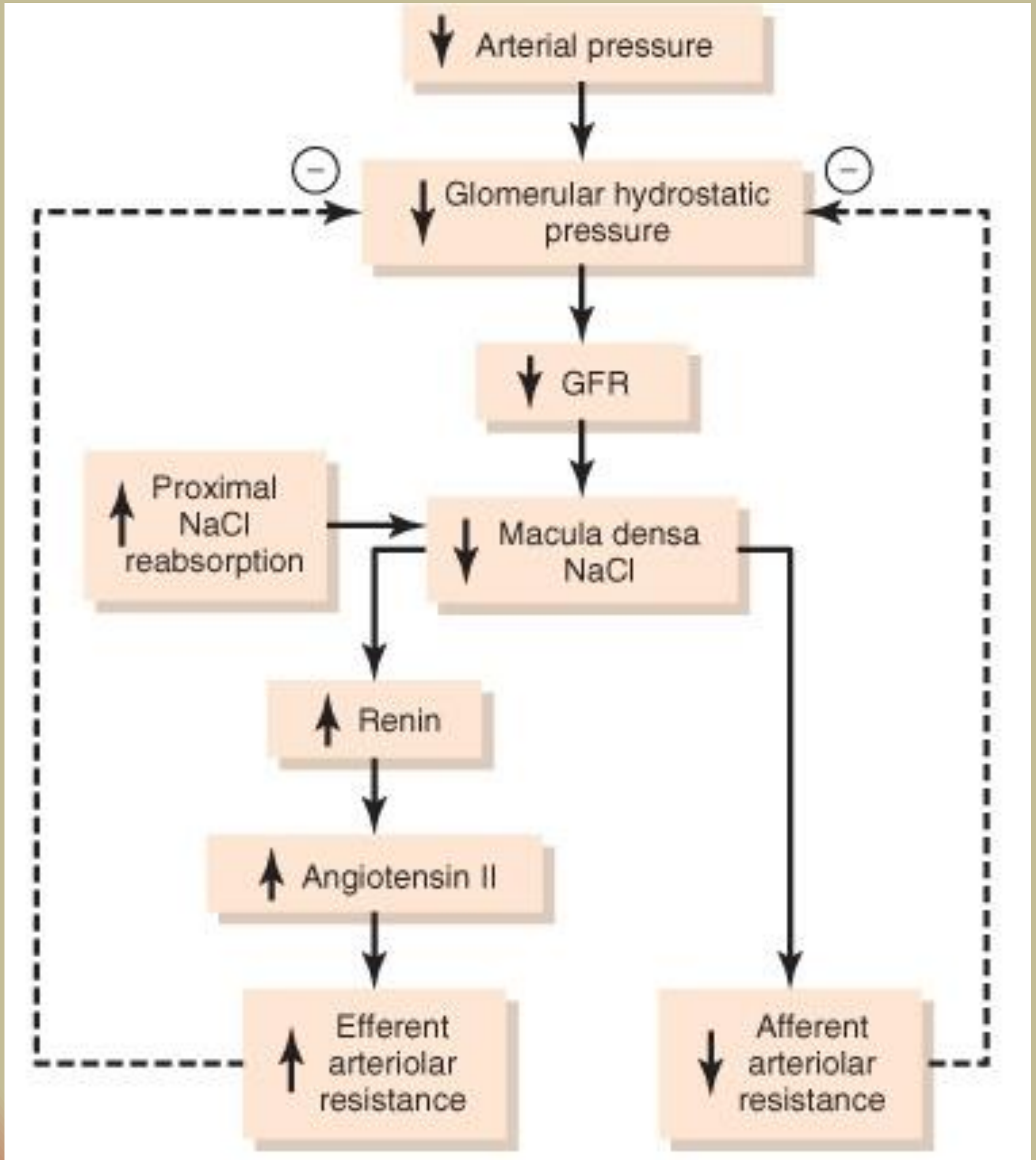
  reduces GFR

- Tubuloglomerular feedback: macula densa provides feedback to glomerulus inhibits release of NO causing afferent arterioles to constrict and decreasing GFR

Regulation of Glomerular Filtration Rate
Renal Auto-regulation

Regulation	Major Stimulus	Mechanism	Effect on GFR
Myogenic	Stretching of afferent arteriole walls due to increased systematic BP	Contraction of smooth muscles in afferent arteriole wall	Decrease GFR by constricting the lumen
	Decline in glomerular blood pressure	Dilation of AA and G. capillaries Constriction of EA	Increases GFR
Tubuloglomerular feedback	Rapid increase in Na+ and Cl- In lumen at the macula densa due to increased BP	Decreased release of Nitric Oxide by JGA causing AA constriction	Decrease GFR and filtrate volume

Tubuloglomerular feedback



Guyton & Hall: Textbook of Medical Physiology 11e - www.studentc

Macula Densa:

- Low blood pressure:
Reduced flow, increases the reabsorption of Na + and Cl- increased renin secretion

- High blood pressure:
increased flow, decreases absorption of NaCl
Reduced renin secretion

Mechanisms regulating GFR

2. Neural regulation
   - Kidney blood vessels supplied by **sympathetic autonomic nervous system (ANS)** fibers that release norepinephrine causing vasoconstriction
   - Moderate stimulation – both afferent and efferent arterioles constrict to same degree and GFR decreases
   - Greater stimulation constricts afferent arterioles more and GFR drops

Regulation of Glomerular Filtration Rate
Neural Regulation

Regulation	Major Stimulus	Mechanism	Effect on GFR
Sympathetic Nerves (Autonomic)	Acute fall in systematic blood pressure. Release of norepinephrine	Constriction of afferent arterioles	Decrease GFR and filtrate volume to maintain blood volume

Mechanisms regulating GFR

3. Hormonal regulation

- **Angiotensin II** reduces GFR
- **Atrial natriuretic peptide** increases GFR
- **Antidiuretic hormone (ADH)**
- **Aldosterone**

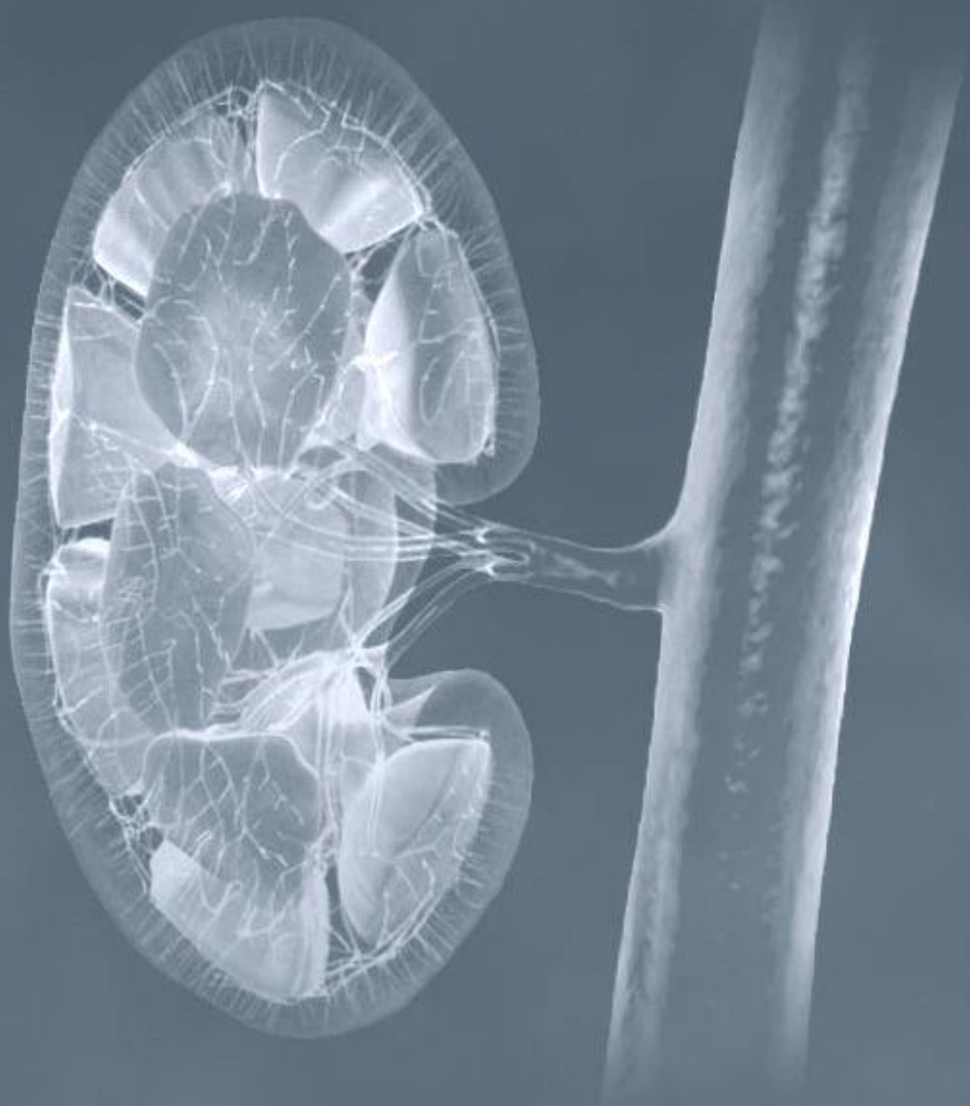
Regulation of Glomerular Filtration Rate
Hormonal Regulation

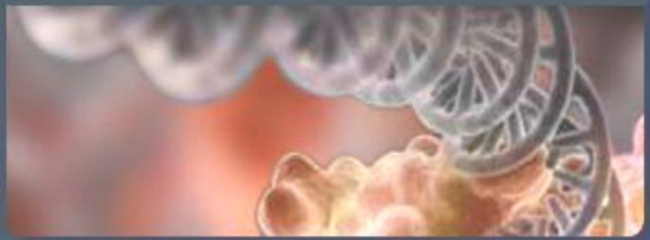
Regulation	Major Stimulus	Mechanism	Effect on GFR
Angiotensin II	Decreased blood volume or decreased blood pressure	Constriction of both afferent and efferent arterioles	Decreases GFR
Atrial natriuretic peptide	Stretching of the arterial walls due to increased blood volume	Relaxation of the mesangial cells increasing filtration surface	Increases GFR

Regulation of Glomerular Filtration Rate
Hormonal Regulation

Regulation	Major Stimulus	Mechanism	Effect on GFR
Antidiuretic hormone - ADH	Increased Angiotensin II or decreased volume of extracellular fluid	Stimulate insertion of aquaporin-2 (water channels) In apical membrane or principal cells	Increases blood volume to return GFR to normal
Aldosterone	Secreted from adrenal cortex because of increased Angiotensin II levels	Increases reabsorption of Na+ and water by principal cells of the DCT collecting duct	Increases blood volume to return GFR to normal

Kidneys And Blood Pressure Regulation





How the body regulates the blood pressure ?

Two major parameters of Blood Pressure control

1. Total peripheral resistance
2. Cardiac output

Arterial Blood Pressure

= Cardiac Output × Total Peripheral Resistance

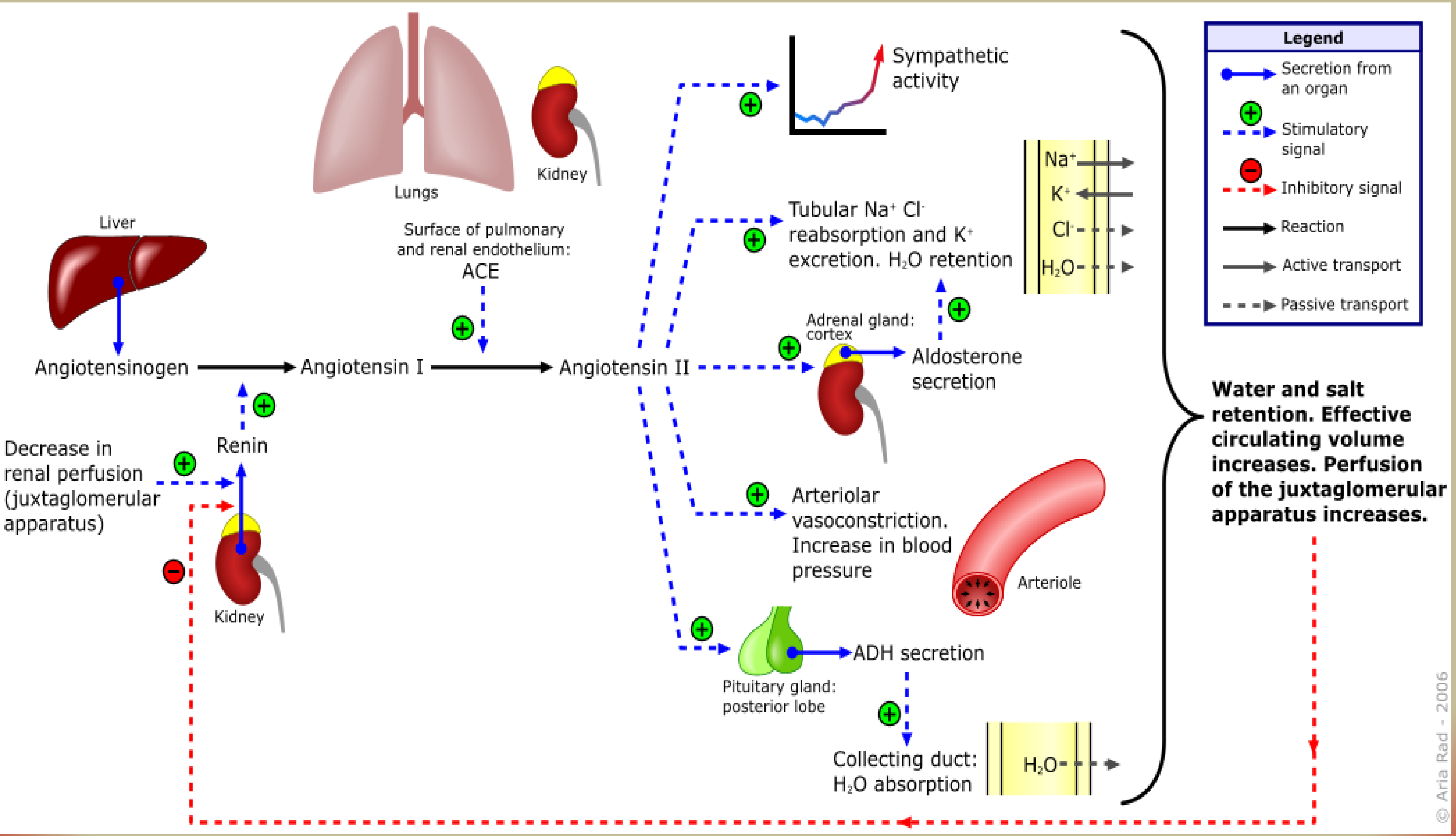
Mechanisms of controlling Blood Pressure

- ***Rapid Control***
  - Baroreceptor
  - CNS ischemic mechanism
  - Chemoreceptors

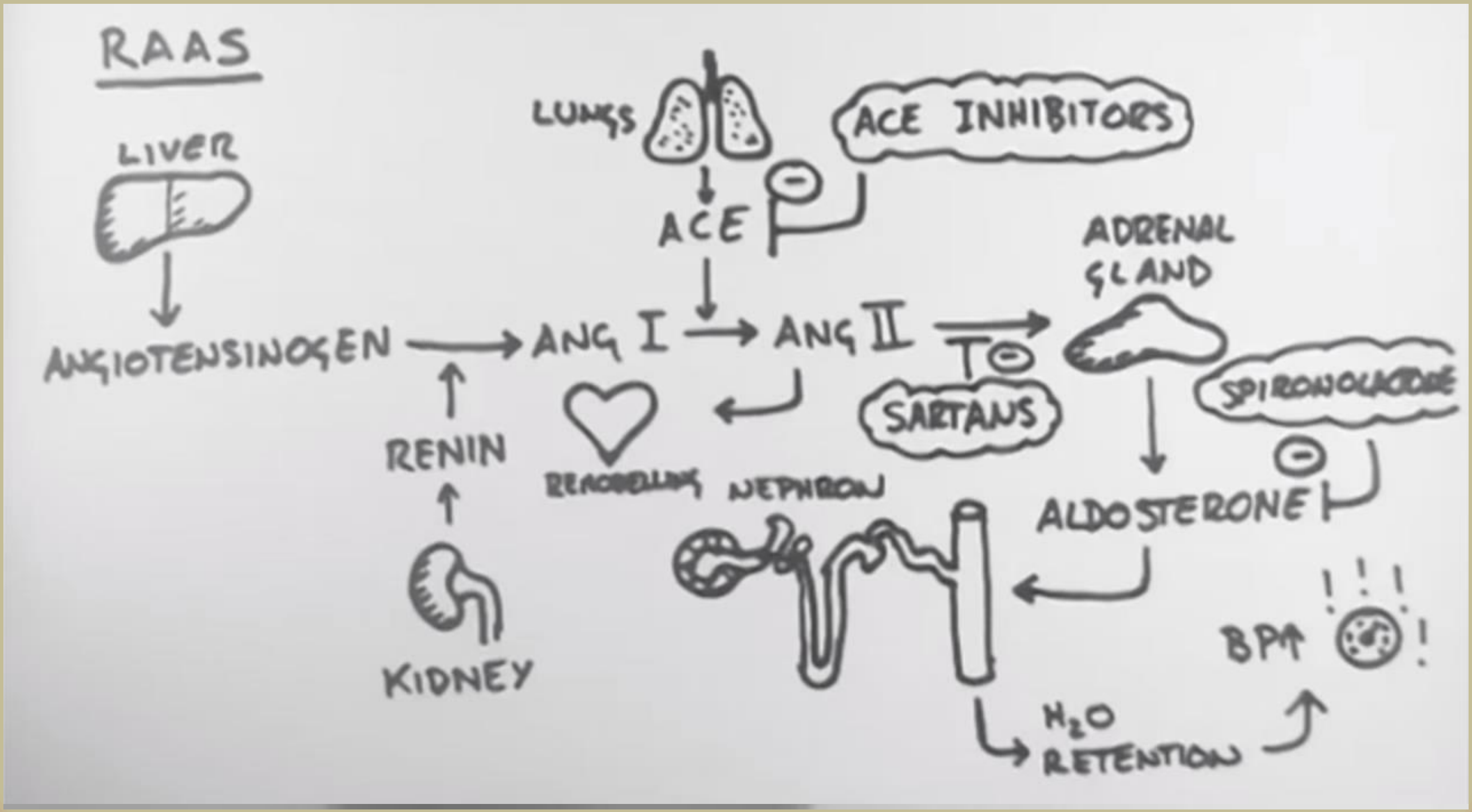
- ***Intermediate Control***

- ***Long-Term Control***
  - Renal-body fluid pressure control mechanism - hours to days
  - RAAS interaction with aldosterone – Regulating extracellular (ECF) volume

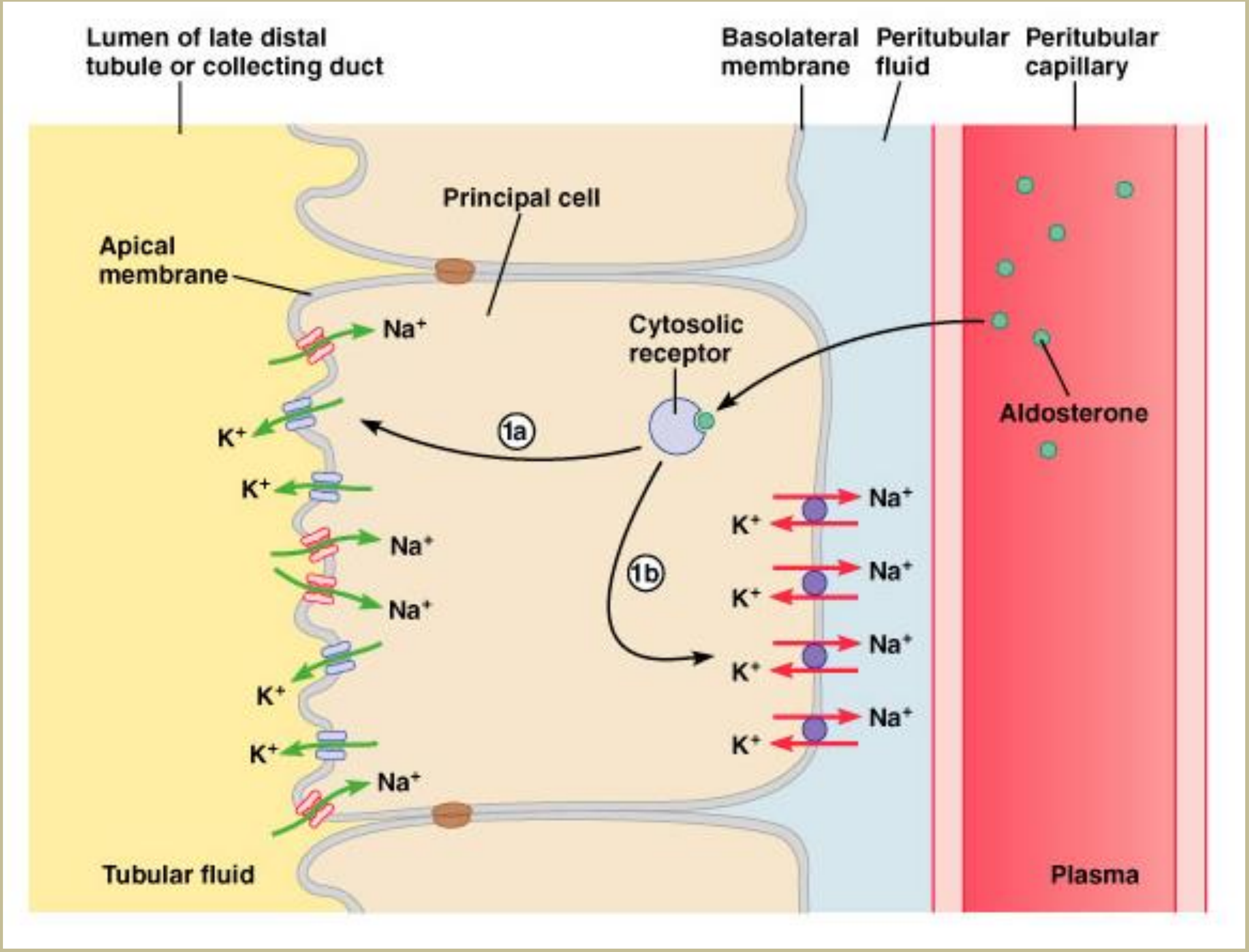
Renin Angiotensin Aldosterone System (RAAS)



Renin-Angiotensin-Aldosterone System



Action of Aldosterone



How RAAS affect blood pressure regulation ?

Increase in Blood volume

↓

Increase in venous return

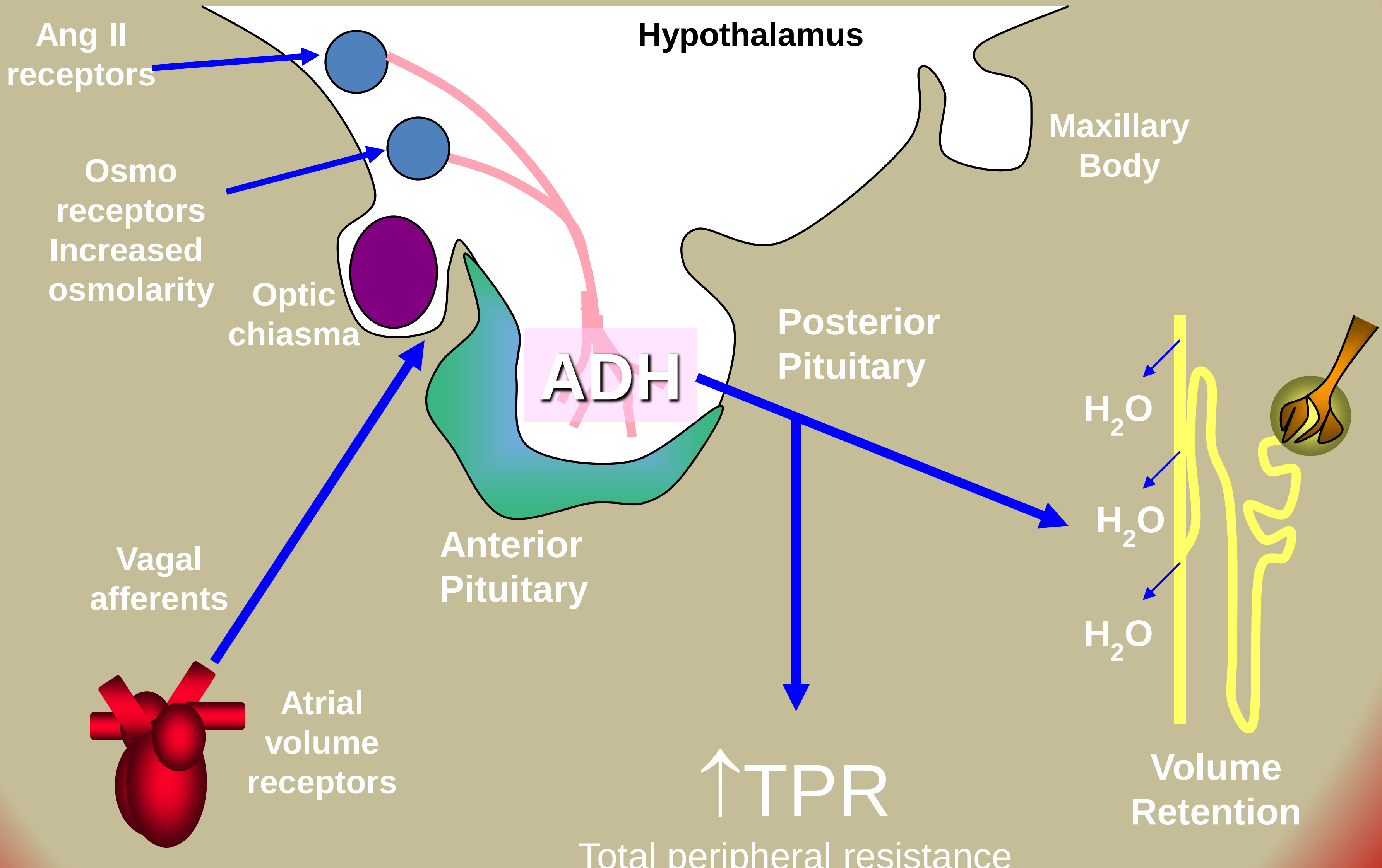
↓

Increase in cardiac output

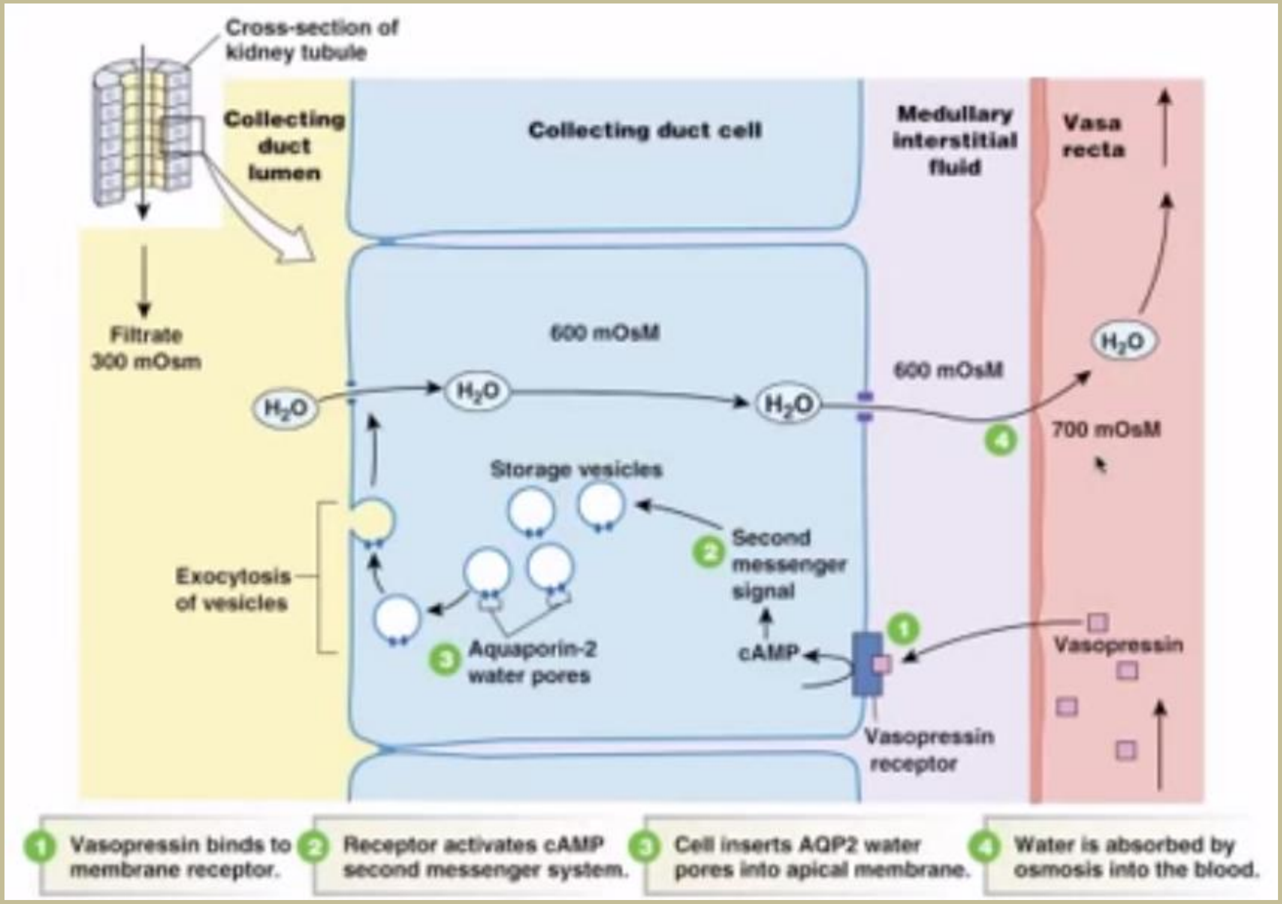
↓

Increase in blood pressure

ADH (Vasopressin) and Blood Volume



Action of ADH (Vasopressin)



Arterial Volume receptors and Hypothalamic osmoreceptors

Increase pressure due to
Increase in blood volume

↓

Distension of the atrial walls

↓

Stimulation of receptors

↓

Dilatation of renal arterioles

↓

More filtration of fluid
in the kidneys

Decrease osmotic pressure

↓

Hypothalamus

↓

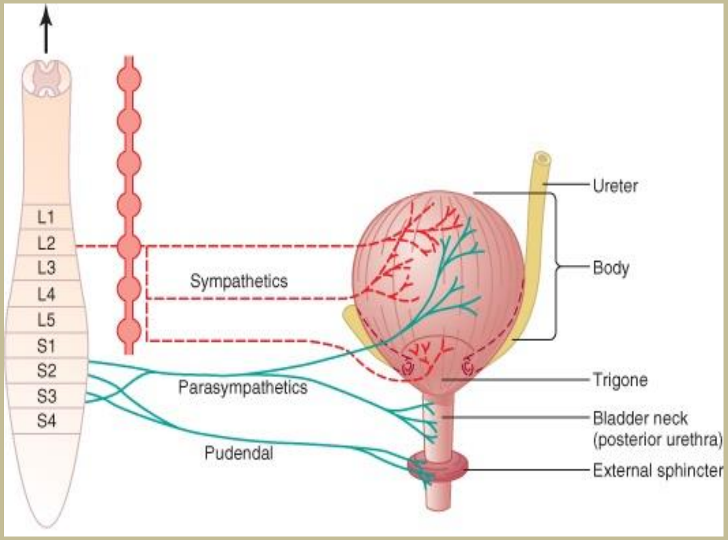
Decrease ADH secretion

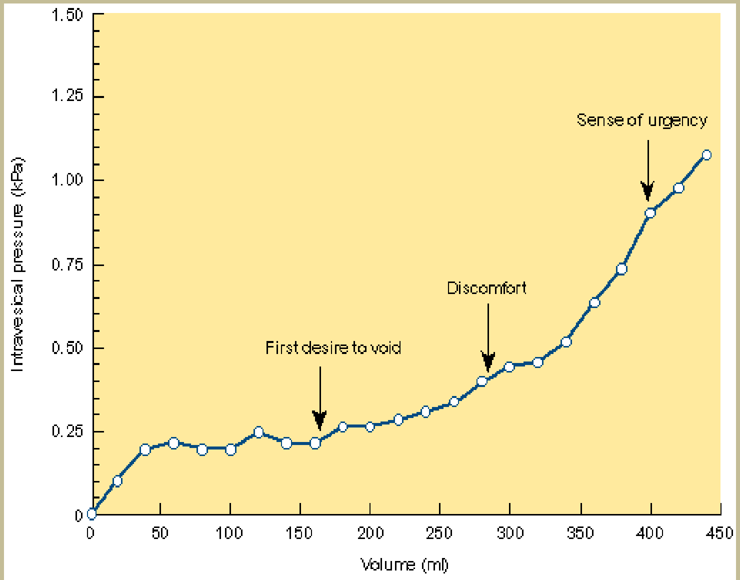
↓

Decrease water reabsorption
from the kidney

Bring the Volume and BP back to normal

MICTURITION

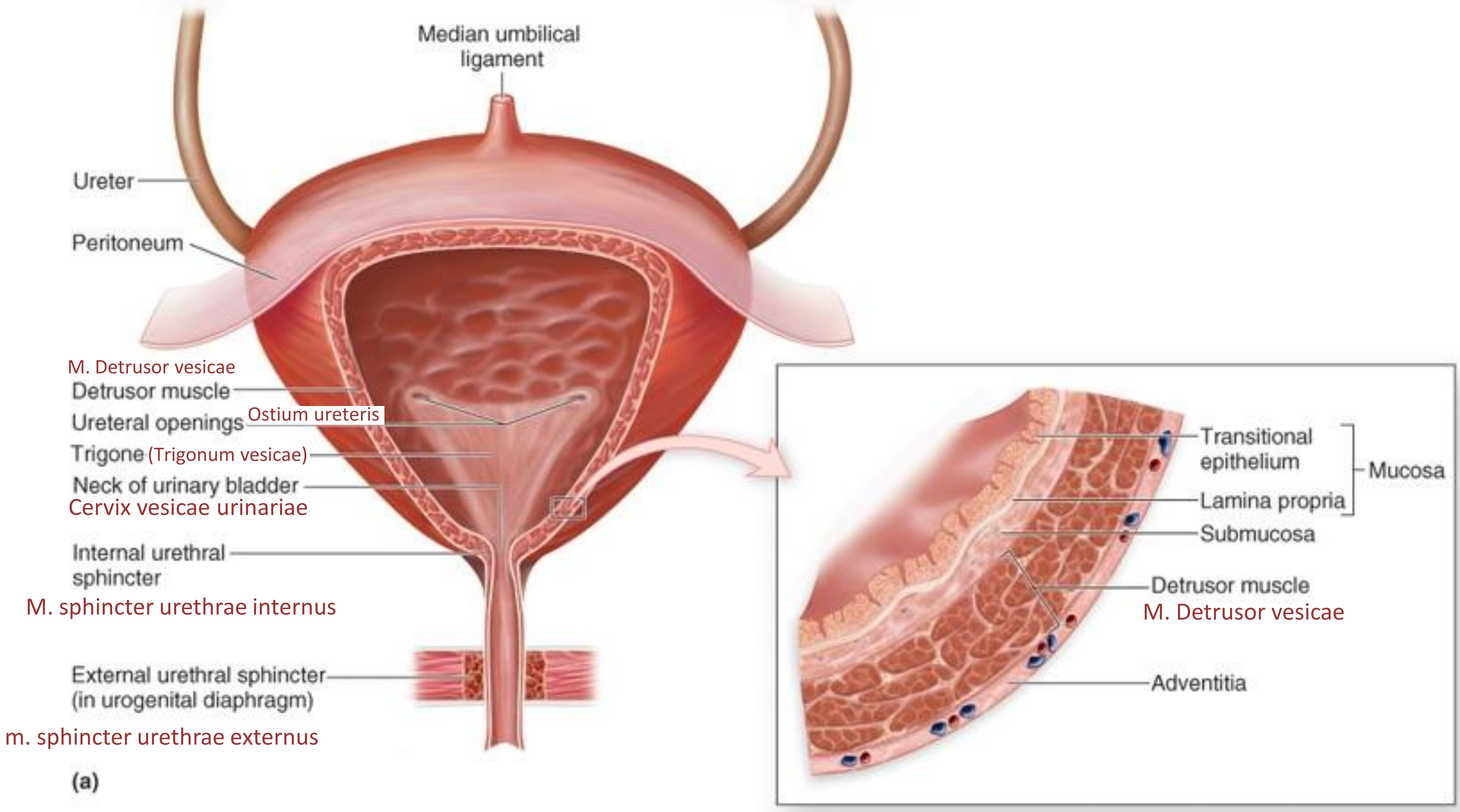




MICTURITION

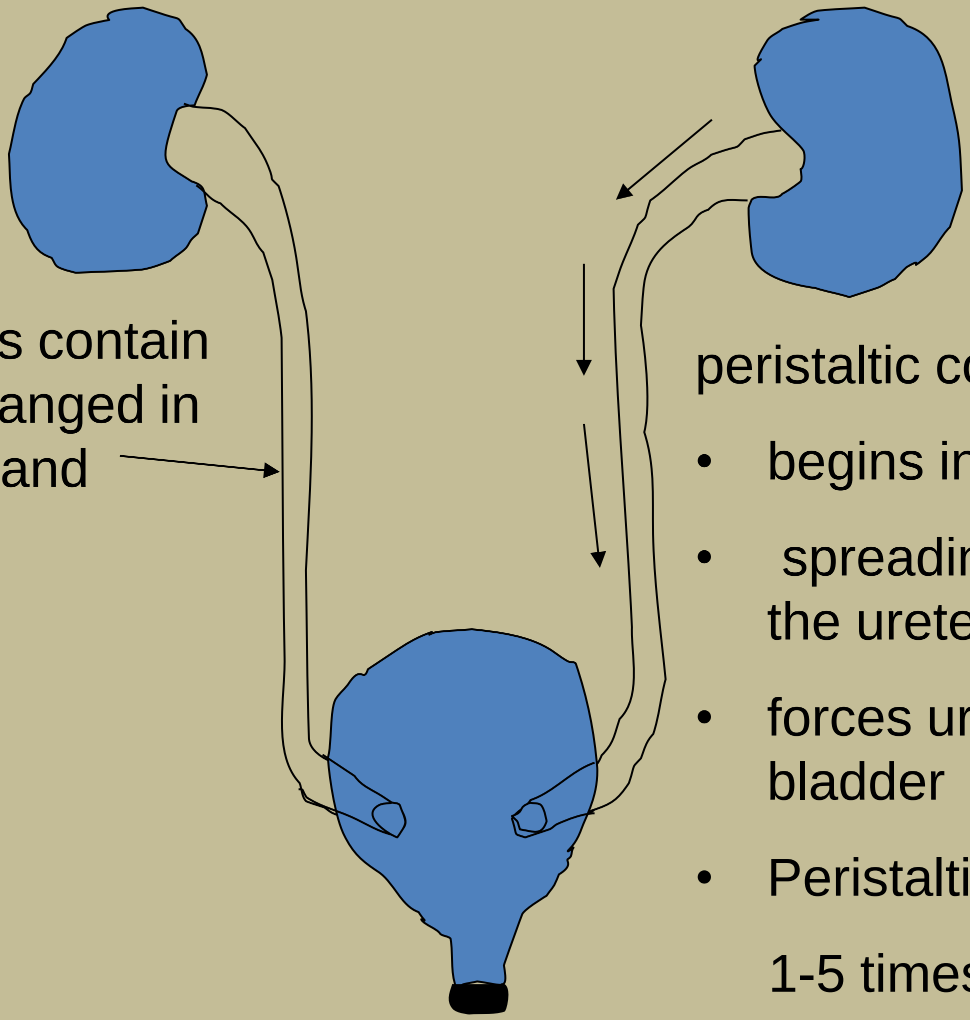
The process by which the urinary bladder empties when it becomes filled

Urinary Bladder



(a)

Filling of the bladder



The walls of ureters contain smooth muscle arranged in spiral, longitudinal and circular bundles.

peristaltic contraction:

- begins in the pelvis
- spreading downward along the ureter
- forces urine toward the bladder
- Peristaltic waves occur

  1-5 times/minute

INNERVATION OF BLADDER

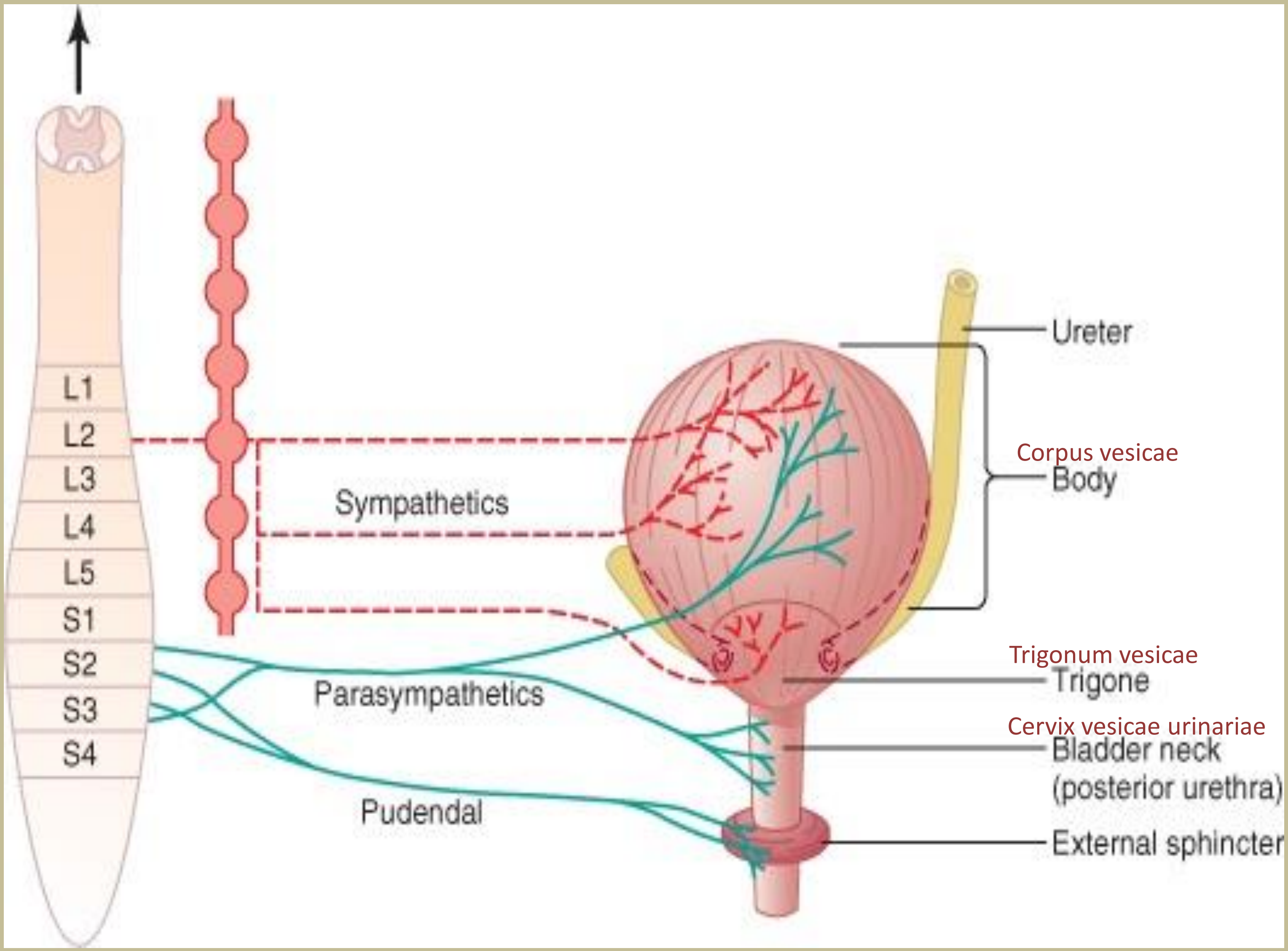
1. **PARASYMPATHETIC NERVES** (PELVIC NERVE) (S_{2-3})
   a) Sensory (stretch)
   b) Motor (detrusor contraction, Internal sphincter)

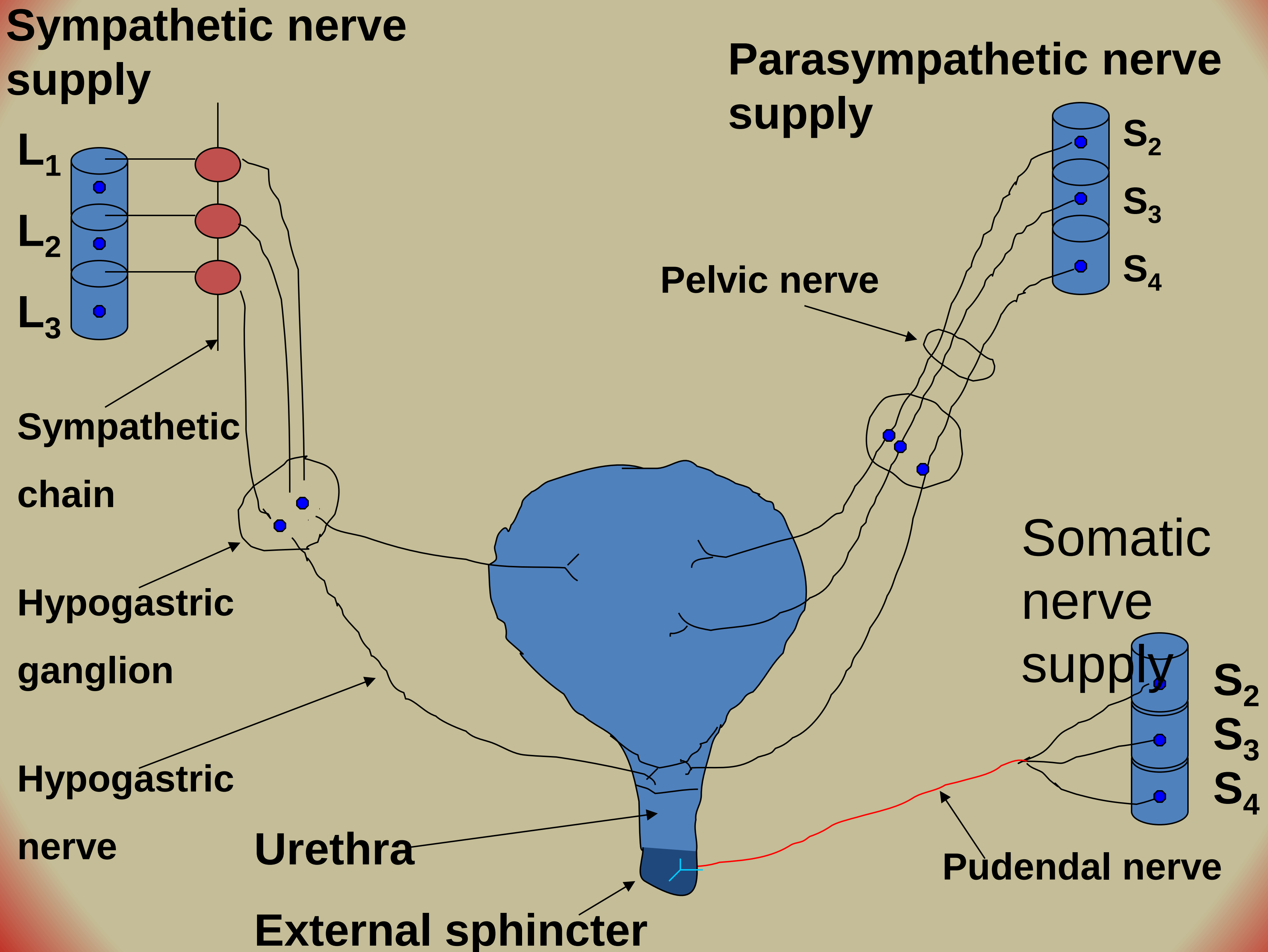
2. **SKELETAL MOTOR FIBER** (PUDENDAL NERVES) (S_{2-3})
   a) Sensory (stretch)
   b) Motor (external sphincter)

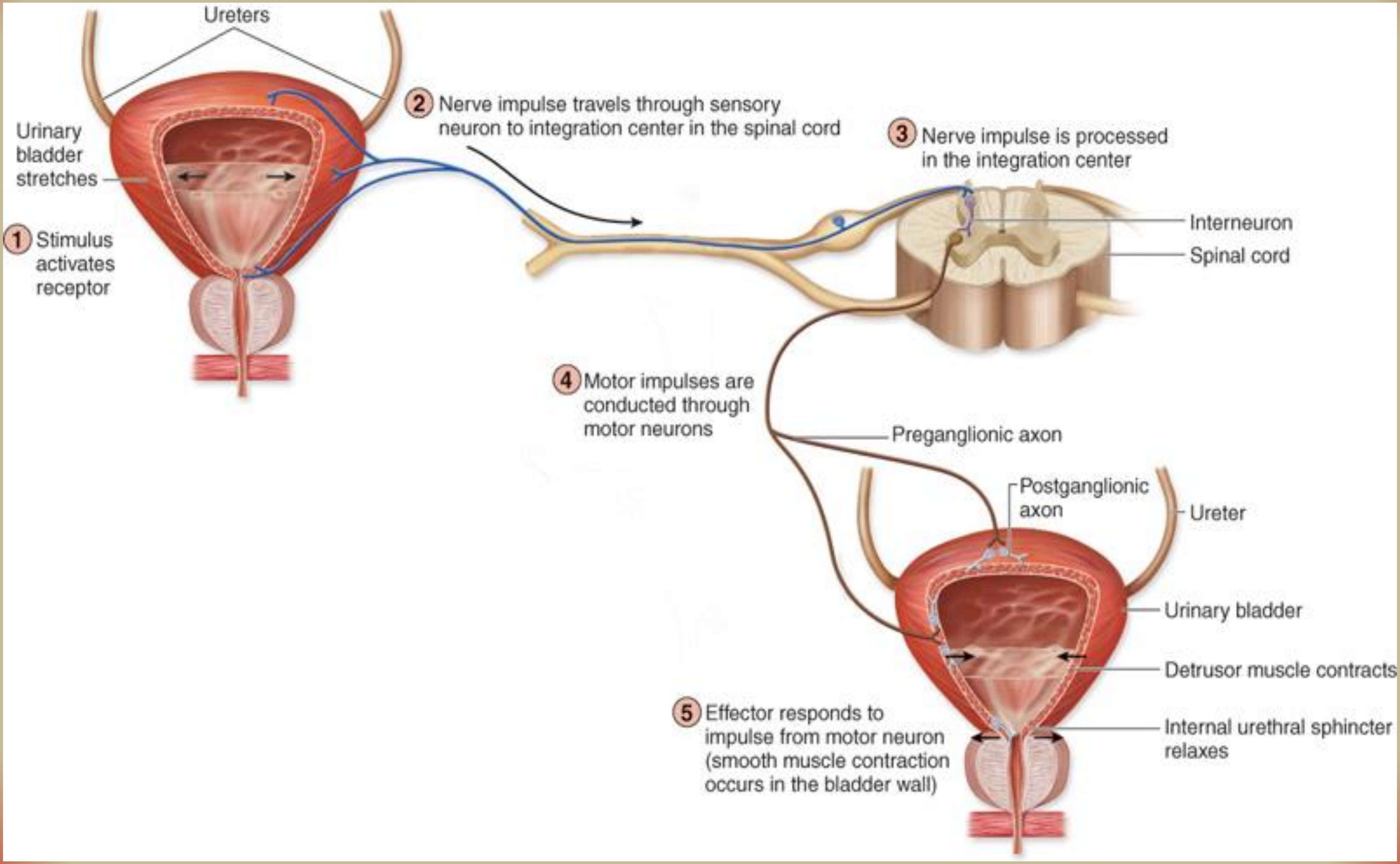
3. **SYMPATHETIC NERVES** (HYOGASTRIC NERVES) (L_2)
   a) Sensory (fullness, pain)
   b) Motor (detrusor relaxation, stimulate blood Vessels)

They also prevent reflux of semen into the bladder during ejaculation (retrograde ejaculation).

Bladder Innervation

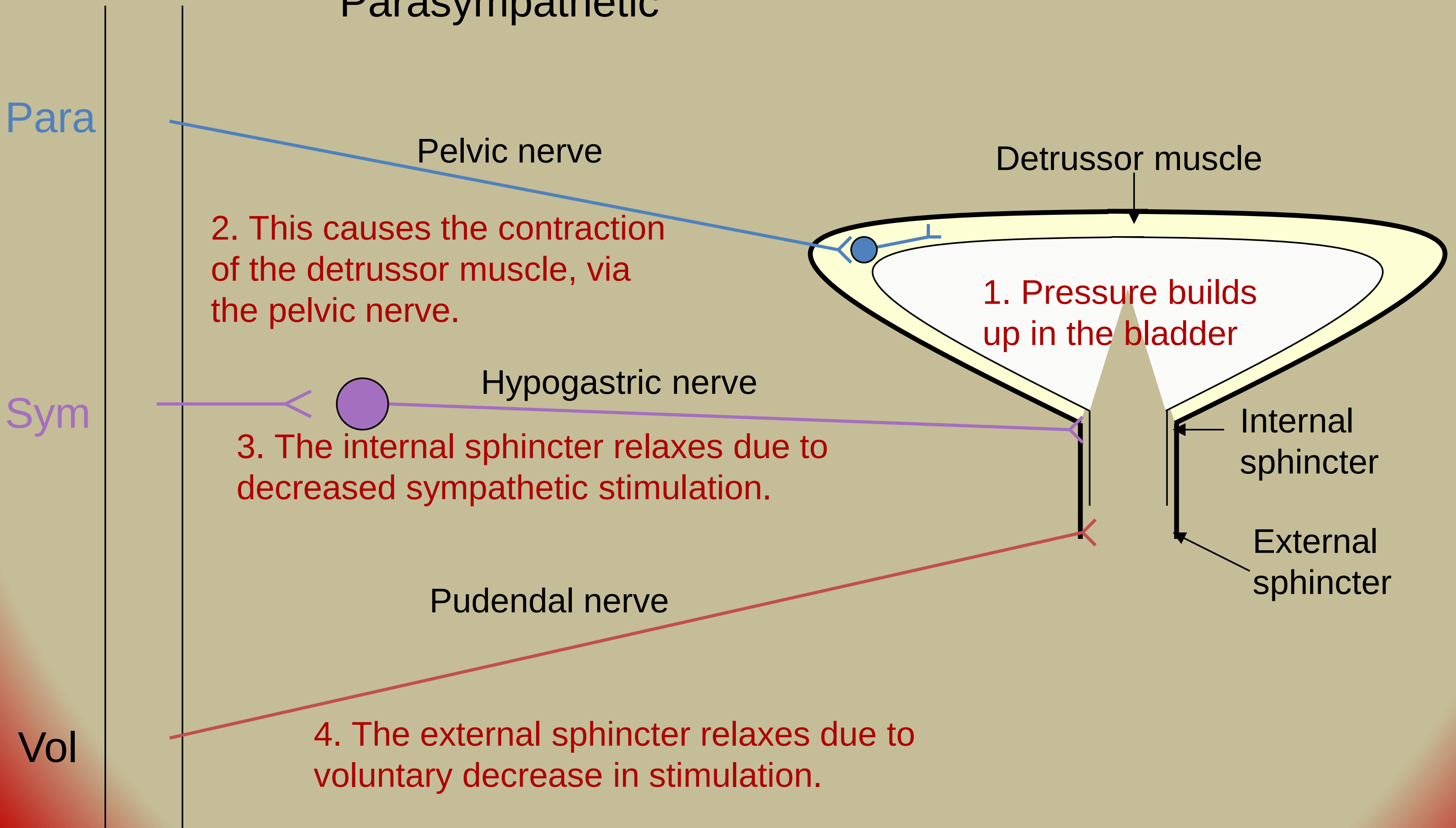



Sympathetic nerve supply
L_1
L_2
L_3
Sympathetic chain
Hypogastric ganglion
Hypogastric nerve
Parasympathetic nerve supply
S_2
S_3
S_4
Pelvic nerve
Somatic nerve supply
S_2
S_3
S_4
Pudendal nerve
Urethra
External sphincter

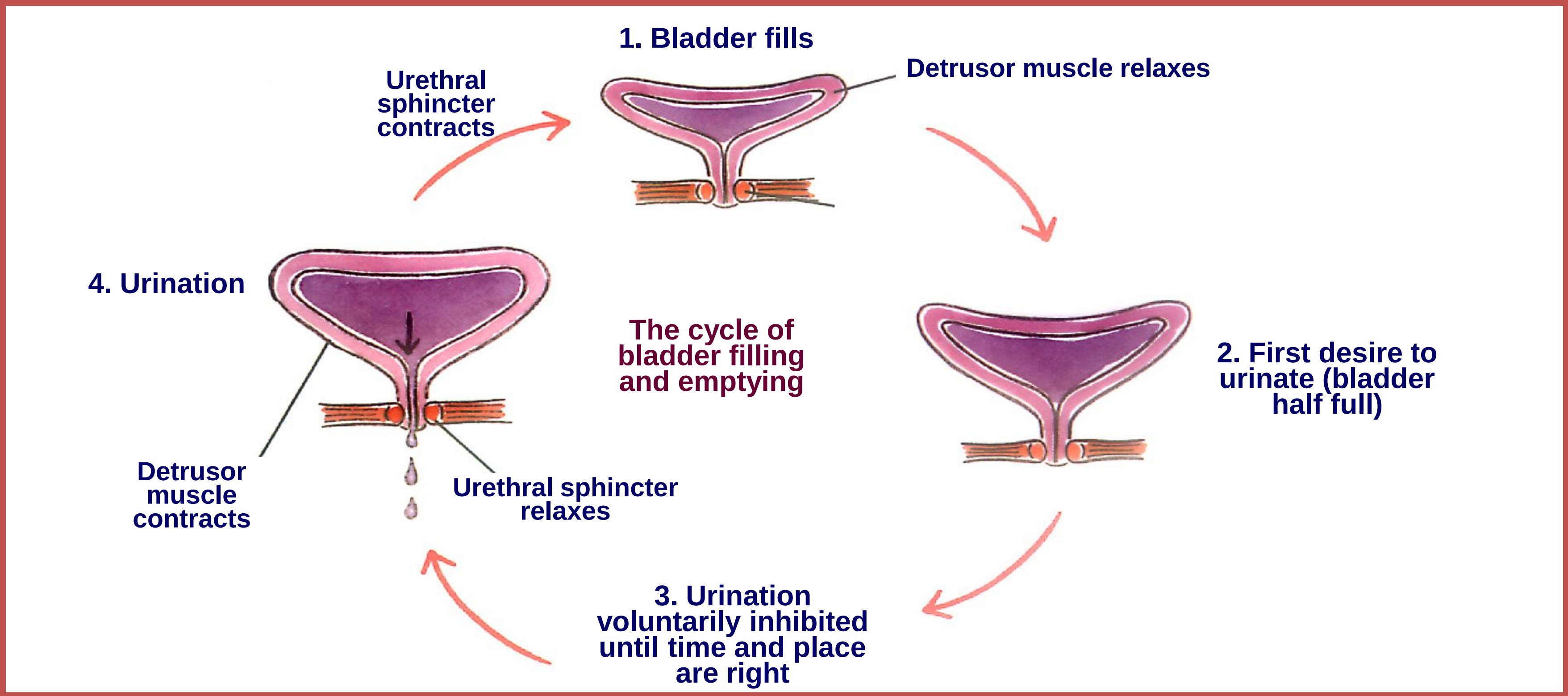

Ureters
Urinary bladder stretches
1 Stimulus activates receptor
2 Nerve impulse travels through sensory neuron to integration center in the spinal cord
3 Nerve impulse is processed in the integration center
Interneuron
Spinal cord
4 Motor impulses are conducted through motor neurons
Preganglionic axon
Postganglionic axon
Ureter
Urinary bladder
Detrusor muscle contracts
5 Effector responds to impulse from motor neuron (smooth muscle contraction occurs in the bladder wall)
Internal urethral sphincter relaxes

Regulation of the Bladder

Main Influence:
Parasympathetic



Bladder Filling & Emptying Cycle

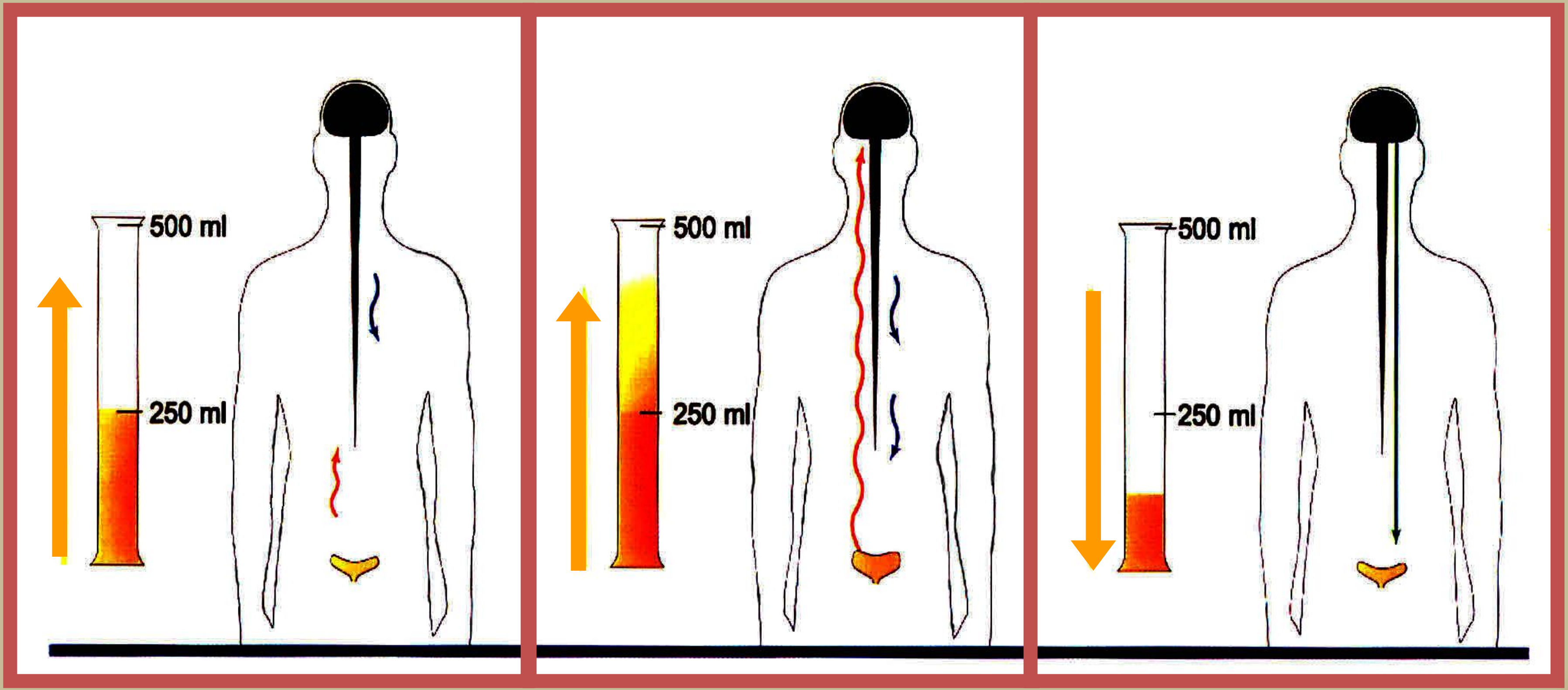


Micturition Reflex

- Micturition contractions begin
  - Role of sensory and motor parasympathetic nerves

- Self regenerative
  - Rapid increase in pressure
  - Period of sustained pressure
  - Return to basal tone

Normal Control of Urination



Voiding Urine - Micturition

- Micturition reflex

1) 300-400 ml urine in bladder, stretch receptors send signal to spinal cord (S2, S3)

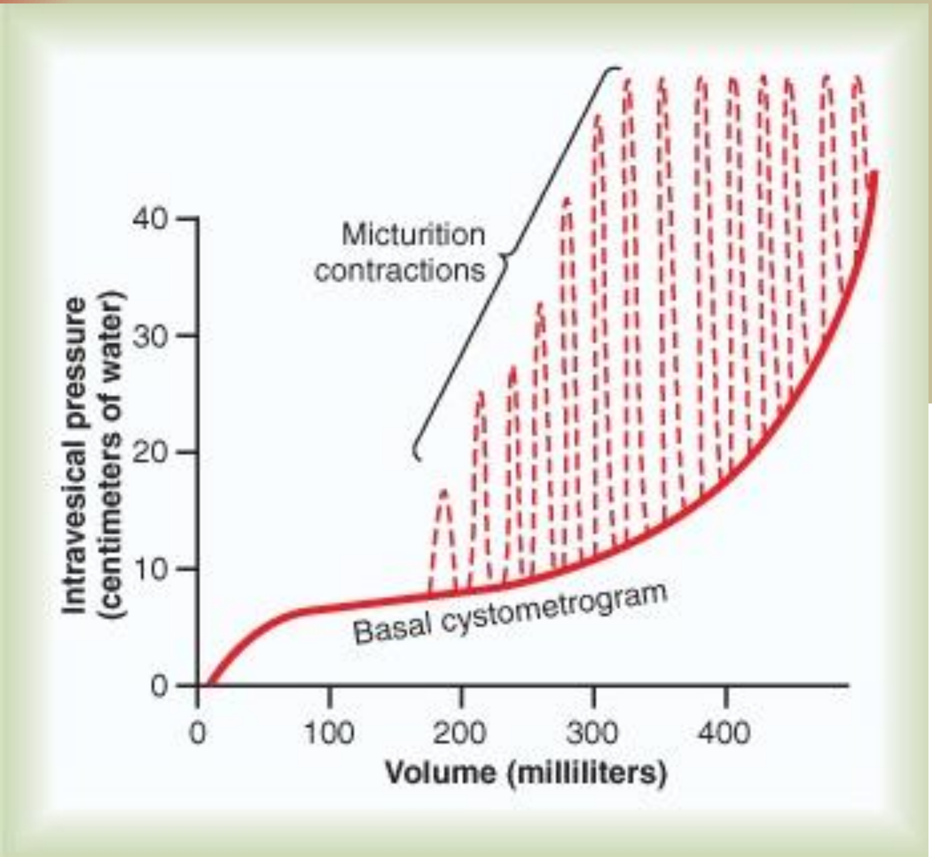
2) parasympathetic reflex arc from spinal cord, stimulates contraction of detrusor muscle

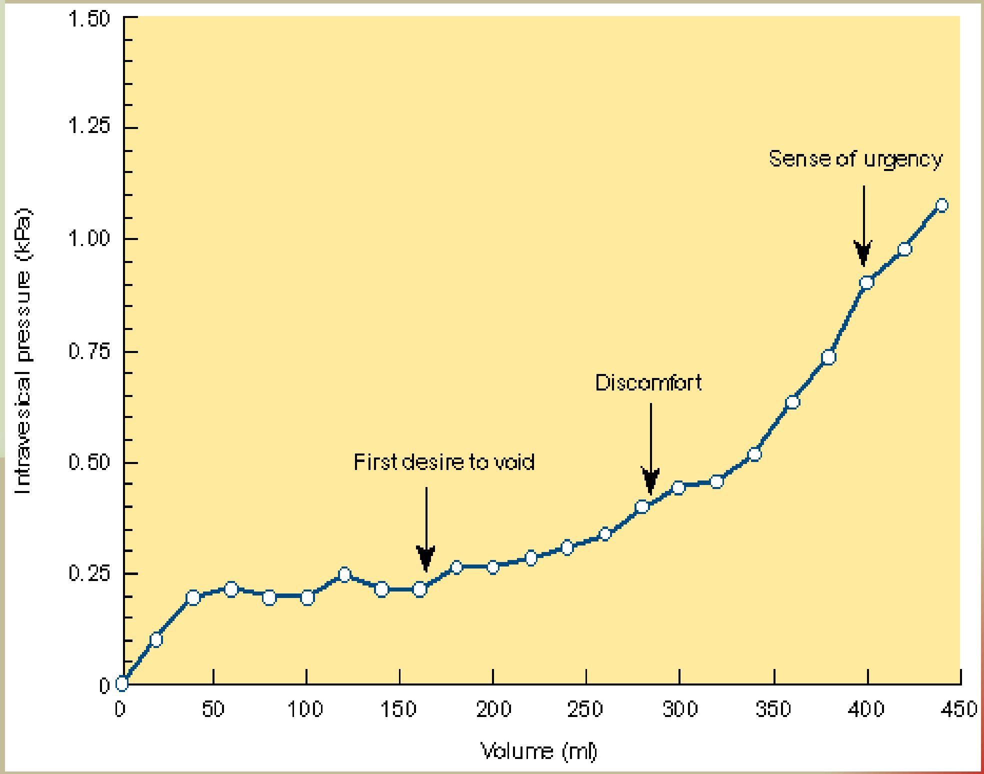
3) relaxation of internal urethral sphincter

4) Voluntary control is possible until 600 – 700 mL of urine
   This reflex predominates in infants

5) Beyond 600 – 700 mL of urine voluntary control starts failing.

Cystometrogram





Bladder filling – **Cystometrogram**

- Relation between bladder volume & pressure.
- Empty bladder……P zero
- 30-50 ml urine……P 5-10 cm H_2O
- 50 – 300 ml urine…. P 5-10 cm H_2O
- More than 400 ml…..rapid rise in P

LAW of LAPLACE

This is in accordance with law of Laplace.

- In the bladder tension increases as the urine is filled.
- At the same time, the radius also increases due to relaxation of the detrusor muscle.
- Because of this, there is almost no pressure rise.

Facilitation or inhibition of micturition by brain

- **Pons**
  - **Facilitatory and inhibitory centers**
- **Cortex**
  - **Mainly inhibitory centers**
  - **Voluntary Urination**

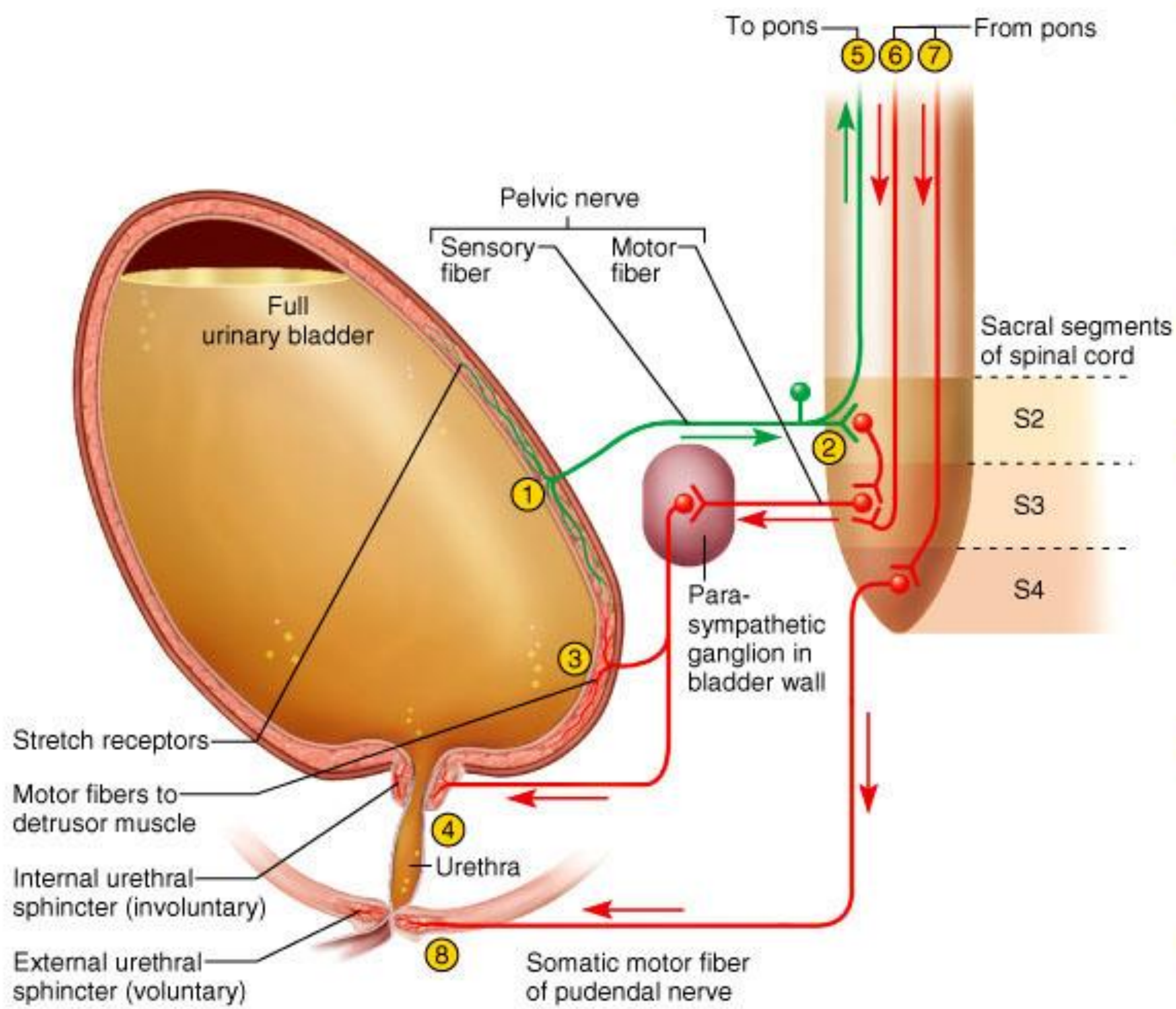
- Micturition center is located in the Frontal lobe
- Function of micturition center:
  - Send tonically inhibitory signals to the detrusor muscle to prevent the bladder from emptying (contracting) until a socially acceptable time and place to urinate is available.

Next stop is the…..

Pons

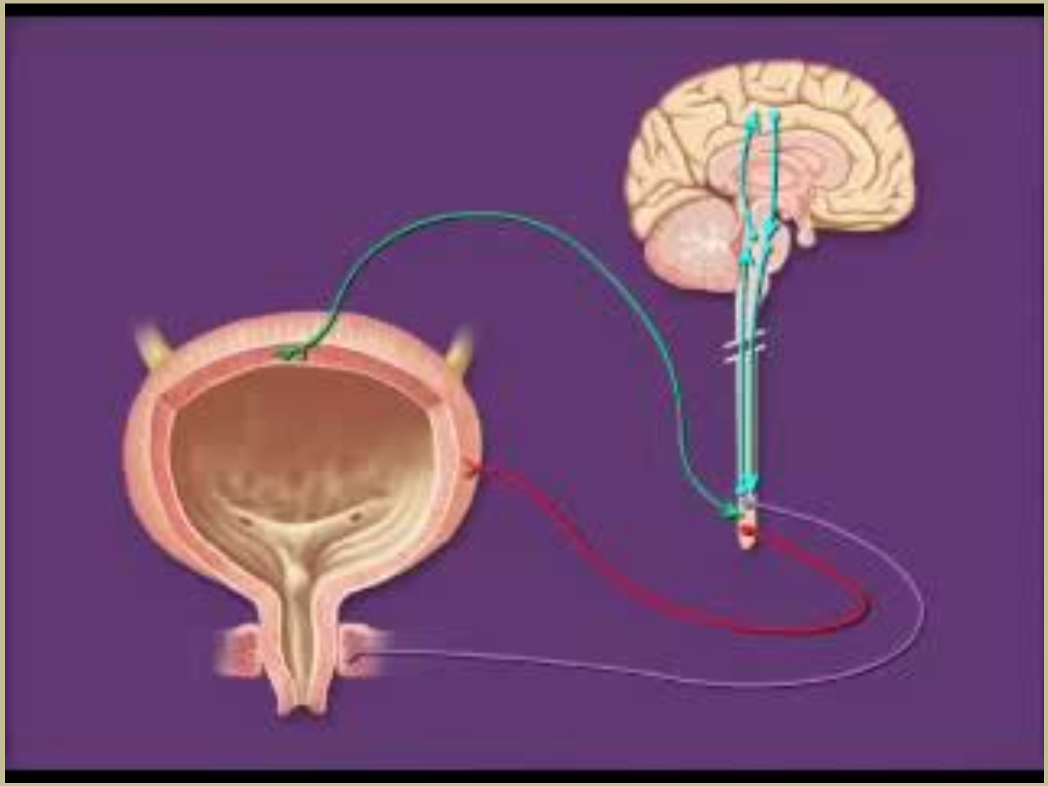
- The major relay center between the brain and the bladder

- Pontine micturition center (PMC):
  - coordinates the urethral sphincter relaxation and detrusor contraction to facilitate urination

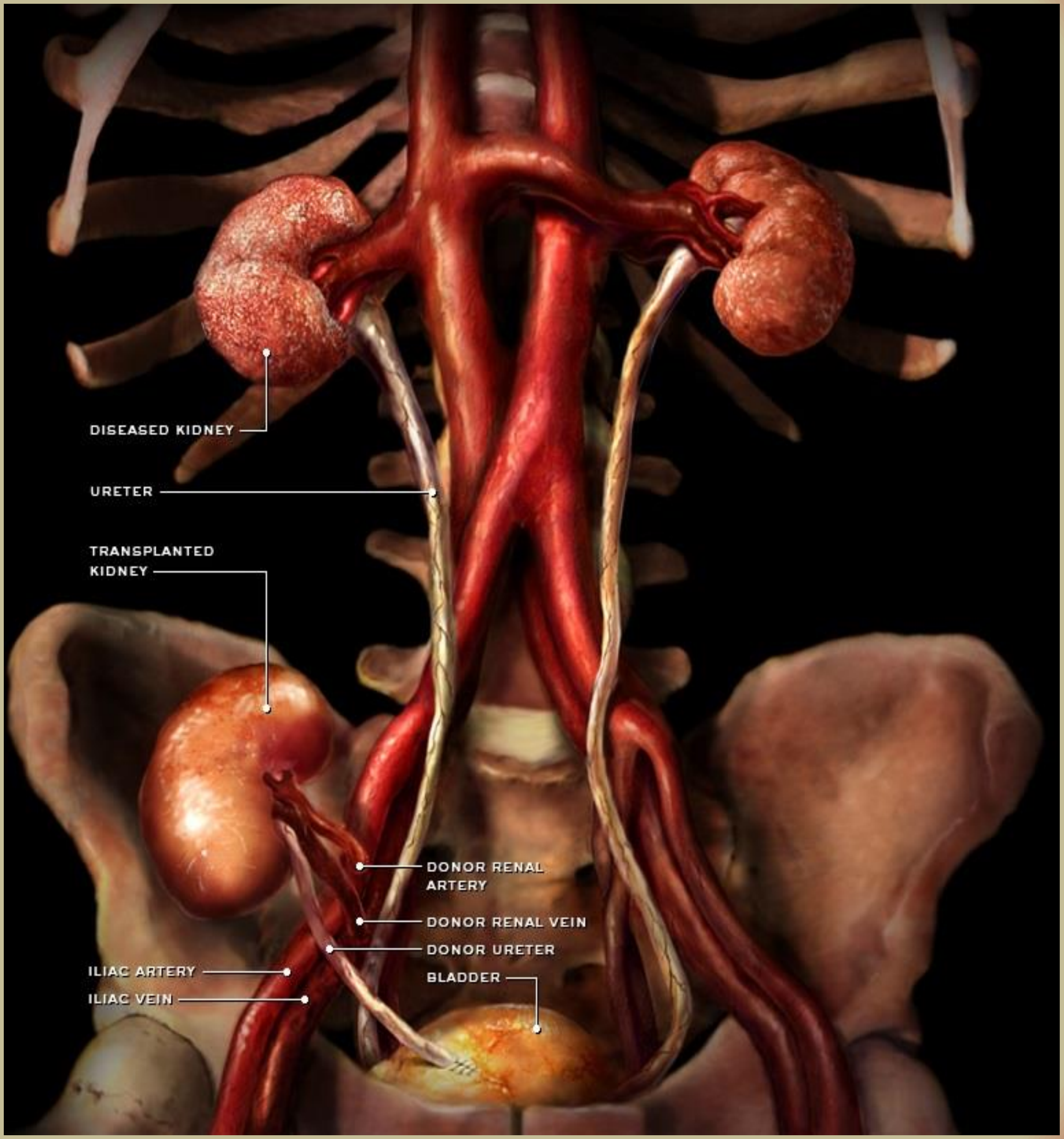


1. Stretch receptors detect filling of bladder, transmit afferent signals to spinal cord.
2. Signals return to bladder from spinal cord segments S2 and S3 via parasympathetic fibers in pelvic nerve.
3. Efferent signals excite detrusor muscle.
4. Efferent signals relax internal urethral sphincter. Urine is involuntarily voided if not inhibited by brain.
5. For voluntary control, micturition center in pons receives signals from stretch receptors.
6. If it is timely to urinate, pons returns signals to spinal interneurons that excite detrusor and relax internal urethral sphincter. Urine is voided.
7. If it is untimely to urinate, signals from pons excite spinal interneurons that keep external urethral sphincter contracted. Urine is retained in bladder.
8. If it is timely to urinate, signals from pons cease and external urethral sphincter relaxes. Urine is voided.

Micturition



Kidney transplant



Isn’t that enough for one day !?!